

A Four-Factor Model for the Indonesia Stock Market

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Abstract

We propose a four-factor model that includes market, size, value, and profitability factors to explain the cross-section of stocks returns for the Indonesian stock market (IDX). Asset spanning tests identify size (*market capitalization*), value (*cash flow-to-price*), and profitability (*EBIT/NOA*) as the most robust characteristics for forming factors. Our four-factor model (IDX4) outperforms existing factor models in digesting stock market anomalies and explaining variation in returns on equity portfolios sorted by firm characteristics. The expected returns implied from the IDX4 model are close to the realized returns on characteristics sorted portfolios. The IDX4 model also performs well in subsamples including Sharia-compliant listings and listings on different exchange boards.

JEL Classification: G12; G14

Key words: Indonesian stock market; factor models; value premium; profitability effect, stock market anomalies.

We thank K.C. John Wei for helpful comments and suggestions. The factors are available for download at <http://idnfactors.online>. All remaining errors are ours.

A Four-Factor Model for the Indonesia Stock Market

Abstract

We propose a four-factor model that includes market, size, value, and profitability factors to explain the cross-section of stocks returns for the Indonesian stock market (IDX). Asset spanning tests identify size (*market capitalization*), value (*cash flow-to-price*), and profitability (*EBIT/NOA*) as the most robust characteristics for forming factors. Our four-factor model (IDX4) outperforms existing factor models in digesting stock market anomalies and explaining variation in returns on equity portfolios sorted by firm characteristics. The expected returns implied from the IDX4 model are close to the realized returns on characteristics sorted portfolios. The IDX4 model also performs well in subsamples including Sharia-compliant listings and listings on different exchange boards.

1. Introduction

The Indonesian Stock Exchange (IDX) is one of the fastest growing equity markets in the world. From 1990 to 2023, the IDX index experienced an annualized return of 10.01% in local currency terms. Indonesia provides a rich setting for studying important finance topics such as financial crises, liquidity risk, and investor sophistication (Mitton (2002), Bekaert, Harvey, and Lundblad (2007), Dvořák (2005), Rhee and Wang (2009)). Our aim is to develop a factor model that best captures the expected returns of stocks traded on the IDX because existing factor models provide no improvement in describing cross-sectional stock returns in Indonesia (Foye and Valentinčič (2020)). Given that researchers draw economic conclusions based on stock performance, a robust factor model would provide a useful benchmark for academics, investors, and practitioners to conduct event studies, assess risk premiums, and evaluate fund performance.

We design a new factor model called the IDX4. The IDX4 comprises of four factors: *market*, *size*, *value*, and *profitability*. Our model is guided by two motivating factors. First, Li, Wei, and Zhang (2023) find that the size, value, and profitability clusters are the key risk factors in the Indonesian market. Second, given that the relative importance of anomalies can vary across markets, we hypothesize that a factor model specifically tailored for an individual market will likely outperform off-the-shelf models. We horserace the IDX4 against the leading factors models: the capital asset pricing model (CAPM), the Fama and French (1993) three factor model (FF3), the Carhart (1997) four factor model (CAR4), the q-factor model (Q4) of Hou, Xue, and Zhang (2015), and the Fama and French (2015) five factor model (FF5). Our evidence indicates that the IDX4 model best explains the expected returns of stocks listed on the IDX among all the candidate models.

To construct the IDX4, we first create a set of candidate factor-mimicking portfolios based on size, value, and profitability characteristics. We create a single size factor (small minus big, *SMB*) constructed as the difference in returns between a portfolio of small market capitalization stocks and a portfolio of high market capitalization stocks. The value factors (high minus low, *HML*) are constructed from the difference in returns between a portfolio of value stocks and a portfolio of growth stocks. The profitability factors (robust minus weak, *RMW*) are constructed from the return spread between a portfolio of ‘robust stocks’ with high profitability and a portfolio of ‘weak stocks’ with low profitability.

Using asset spanning tests, we select the most robust characteristic to build value and profitability factors. Our evidence indicates that the best performing value factor is a cash flow-to-price (CP) measure constructed from *operating cash flow-to-price* (OCF/ME). The best performing profitability factor is constructed from *earnings before interest and tax-to-net operating assets* (EBIT/NOA). The CP value factor and the profitability factor subsume other candidate value and profitability factors, but the converse does not hold. They also generate the most efficient factors because they produce the highest Sharpe ratio among all the candidate factors. To formulate the IDX4 model, we follow Fama and French (2015) to orthogonalize the size, value, and profitability factors and add the CAPM market factor.

We analyze the ability of the IDX4 and other factor models to explain the return predictability of characteristics-sorted portfolios. Following Jensen, Kelly, and Pedersen (2023), we construct 149 long-short portfolios of possible stock return predictors. 32 of the long-short (LS) strategies earn significantly positive average returns during our sample period. Our four-factor model IDX4 explains 29 out of the 32 significant LS strategies, the most among all models

examined. In comparison, the FF5 and Q4 explain the average returns on 15 and 20 significant LS strategies, respectively. The CAPM explains 14 of the significant LS strategies.

The IDX4 model also produces expected returns closest to the average realized returns of 5x5 size-by-value and 5x5 size-by-profitability portfolios. To visualize the patterns, we plot the IDX4-implied portfolio expected returns against the realized portfolio returns. The scatterplot shows that the IDX4 model closely fits the realized returns on characteristic-sorted portfolios because the data points are located around the 45-degree line. The premiums on small stocks with high valuation ratios or high profitability remain a challenge to the IDX4.¹ Overall, the results suggest that the IDX4 model out-performs off-the-shelf factor models that were originally fitted for the U.S. market.

We also assess the ability of the IDX4 model to explain the time-series variation in stock returns. First, we compute the absolute mean alpha of the 32 significant long-short strategies. The IDX4 and its 3-factor versions are the most successful at explaining the time series variation of portfolios sorted by firm characteristics. Second, we conduct Gibbons, Ross, and Shanken (1989) full-sample multivariate tests (henceforth, GRS). The IDX4 model outperforms other factor models in producing the most insignificant GRS p-values across the 32 test portfolios.

Our final analysis exploits special features of the Indonesian stock market to evaluate the performance of the IDX4. Over 80% of the population in Indonesia identify as Muslim. Islamic tradition includes a religious law call Sharia, which contains a set of rules that govern banking and finance. The IDX identifies stock listings that are compliant with Sharia principles. Firms adhering to the Islamic laws exhibit different risk management (Majdoub and Sassi (2017)) and dividend payout policies (Ben-Nasr and Ghouma (2022)). Domestic investors may categorize the two types

¹ In the U.S. market, these portfolios also pose a challenge to extant factor models (Fama and French (1993), Fama and French (2015)).

of stocks differently and evaluate them in different ways. First, we estimate LS strategies separately among Sharia and non-Sharia stocks. The value effect and size effect exist in both samples. However, the profitability effects only exist among non-Sharia stock. Of all the factor models, IDX4 explains the most anomalies within both subsamples.

Next, we separately analyze stocks listed on the two listing boards of the IDX. The Main board lists larger stocks with long trading history and active trading. The Development or Acceleration board lists smaller stocks with short trading history and those with high growth potential. We were surprised to find that more significant anomalies exist on the main board, which indicates that the size effect is unlikely to explain the predictability of many firm characteristics. The IDX4 again performs the best in explaining average returns on long-short portfolios sorted by firm characteristics within each subsample.

Our paper adds to the growing research on the Indonesian financial market. Due to its unique market structure, social norms, and cultural background, Indonesia is a popular research setting for studying economic questions related to tariffs, foreign direct investments, political relationships, corruption, and financial literacy.² Studies that directly examine equity-related research questions include Leuz and Oberholzergee (2006), Dvořák (2005), Rhee and Wang (2009), Agarwal et al. (2009), Arnold and Javorcik (2009), Hadad et al. (2011), and Shaban and James (2018). We develop a robust factor model that provides researchers a useful benchmark to draw reliable inferences on stock returns on the IDX.

Our comprehensive analysis of the cross-section of stock returns in Indonesia relates to extant studies that examine market efficiency, stock performance around the Covid-19 pandemic, and market sentiment (Andrianto and Mirza (2016), Indrayono (2021), Cheema and Fianto (2024),

² See, for example, Amiti and Konings (2007), Leuz and Oberholzergee (2006), Olken (2007), Olken (2009), Cole, Sampson, and Zia (2011).

Bowe and Domuta (2024)). Sharma et al. (2019) find that capital expenditure is priced in the cross-section. The IDX4 accommodates many anomalies in the Indonesian stock market.³

Finally, we contribute to the research on multi-factor models in international markets (Fama and French, 1998). Recent studies focus on the growing Asian equity markets.⁴ Foye and Valentinčič (2020) extends the FF5 model to the Indonesian market and find the multi-factor model provides no improvement in describing cross-sectional stock returns. They attribute the underperformance of the model to poor accounting quality and opaque financial reporting of Indonesian firms. To our knowledge, no prior study has developed a robust model for the Indonesia market. Given the distinctive market structure, economic and political factors, and investors' profiles unique from the other markets examined by existing literature, Indonesia provides a useful out-of-sample test of the performance of multi-factor models in describing stock returns.

2. Construction and selection of risk factors

2.1. Data and sample

Data on financial statement information of Indonesian public firms are retrieved from Compustat Global Fundamental (Annual and Quarterly). Indonesian stock trading and pricing information are obtained from Compustat Global Security Daily. All raw data are stated in the local currency, the Indonesian Rupiah. We follow the procedures used in Jensen, Kelly, and Pedersen (2023) and Bessembinder et al. (2023) to clean the daily frequency equity market data and calculate monthly stock returns. Following prior studies, we assume that financial statement

³ There is also a related literature documenting that the overall Indonesia stock market performance is affected by economic policy uncertainty, crude oil shocks, macroeconomic factors, and the Covid-19 pandemic (Giral (2019), Herwany et al. (2021), Setiawan (2020)).

⁴ Recent studies on factor models in Asian markets include: China (Guo et al. (2017), Liu, Stambaugh, and Yuan (2019), Li, Liu, Liu, and Wei (2024)), Hong Kong (Bai, Li, and Qin (2017)), India (Balakrishnan, Maiti, and Panda (2018)), Singapore (Hameed, Ni, and Tan (2023)), South Korea (Kim, Kim and Shin (2012)), Taiwan (Chang, Hsieh, and Lai (2009)), Thailand (Sapra and Gupta (2021)), and Vietnam (Huang, Liu, and Shu (2023)).

information become publicly available four months after the fiscal year end dates. The sample period spans from January 1990 through June 2023. We use the 10-year maturity Indonesian government bond as the local risk-free asset.⁵

2.2. Market, size, value, and profitability factors

The academic literature has discovered hundreds of possible return anomalies, which JKP (2023) categorize into 13 distinct clusters. The relative importance of anomalies can vary across markets. We narrow our search to three clusters—size, value, and profitability—because Li, Wei, and Zhang (2023) show that anomalies in these three groups are statistically significant and exhibit sizable return magnitudes. The three groups of factors are theoretically motivated and robust in explaining the cross-section of stock returns in the United States (Fama and French (1993, 2015), Hou, Xue, and Zhang (2015)).

To construct the CAPM market risk factor, we compute the difference each month between the value-weighted average return of all stocks listed on the IDX and the monthly risk-free rate. Banz (1981) documents the size anomaly. We compute the size premium as the difference between the value-weighted returns in largest quintile minus the smallest quintile.

We calculate firm characteristics that measure valuation (17 variables) and firm profitability (11 variables). Detailed definitions of all firm characteristics are available in the Internet Appendix. Then, we use the 2-by-3 procedure of Fama and French (1993) to construct risk factors from equity portfolios that consist of stocks that are sorted by each anomaly variable. Using the book-to-market (BM) ratio as an example, in each month, individual IDX stocks are first sorted by the median firm size of stocks listed in the Main Board into two groups, small and big

⁵ Yields on Indonesian government bond are available starting in 2010. Before that, we use exchange rates between U.S. Dollar and Indonesian Rupiah (obtained from Compustat) to convert yield on 1-month maturity U.S. Treasury Bill into a risk-free rate stated in the local currency.

stocks. Stocks can also be sorted into three subgroups, growth, neutral, and value stocks, by the 30th and 70th percentiles of BM of Main Board stocks.⁶ Altogether, there are six groups as the intersections of size portfolios and BM portfolios. We calculate the value-weighted returns in each of the six portfolios. The BM value factor is the simple average of the two long-Value and short-Growth strategies. The size factor is the simple average of the three long-Small and short-Big portfolios. We follow this procedure to construct value and profitability factors from portfolios sorted by firm size and other value- or profitability-related factors. Following Fama and French (2015), we name size, value and profitability factors as SMB (small minus big), HML (high minus low), and RMW (robust minus weak).

2.3. Average returns of candidate risk factors

Table 1 reports the time series average returns of the market, size, value, and profitability factors. It also reports the corresponding size factor that is orthogonalized to each value or profitability factor. We report the associated t -statistics of each factor to test the null hypothesis that the candidate risk factor earns a zero-mean return.

The size premium exhibits an average monthly return of 1.629% per month and is statistically significant at better than the 1% level (t -stat = 3.01). We observe significantly positive returns at 5% significance level for 13 out of 17 value factors and 7 out of 11 profitability factors. To conserve space, we report only the significant factors. The market portfolio outperforms the risk-free asset by 1.406% per month (t -stat = 3.48). The three value factors with the largest returns exhibit average monthly returns of 2.166%, 1.934%, and 1.754%, and are constructed from *net equity issuance* (*eqnetis_at*), *earnings before interest, taxes, depreciation, and amortization*

⁶ We use Main Board stocks to identify break points to sort portfolios because such stocks have larger market capitalization, longer trading history, higher market liquidity, and less noisy information. Using all stocks to identify break points produces qualitatively similar results.

(*EBITDA*)-to-enterprise value (*EV*) (*ebitda_mev*), and *OCF/ME* (*ocf_me*), respectively. The three measures capture different dimensions of the valuation metric. Net equity issuance tracks financing deficits and surplus. *EBITDA/EV* is perhaps the most widely used valuation multiple in investment banking. *OCF/ME* is a common metric of cash flow-to-price; this measure is the ratio of operation cash flows to the market value of equity.

The three profitability factors with the largest monthly returns earn 1.232%, 1.084%, and 1.006%, respectively, and are constructed from *EBIT/NOA* (*ebit_bev*), *OCF/total assets* (*ocf_at*), and operating profit-to-lagged book equity (*ope_bell*). The three measures capture the concept of return on capital.

The orthogonalized size factor generally exhibits positive average returns but only five are statistically significant at better than the 5% confidence level. Four of the five concentrate in the profitability anomaly. The value effect appears to subsume the size effect for stocks listed on the IDX.

To assess whether the candidate factors contain similar information on expected returns, we calculate pairwise correlations between the factors. Within the same cluster, candidate factors tend to be positively correlated. We observe greater within-group correlations among profitability factors than value factors. Across profitability and value factors, we observe weak or negative correlations in most cases. This result indicates that the two clusters are likely to contain unique information about expected returns. The results are available in Table IA1 of the Internet Appendix.

Overall, the value and profitability factors constructed from various characteristics generally exhibit significant return premiums during our sample period.

2.4. Selecting factors

To select robust factors, we use two asset pricing techniques. First, we rank the Sharpe ratios of candidate three-factor models that include the market, size, and one of the value or profitability factors. Barillas and Shanken (2017) and Fama and French (2018) show that it is sufficient to examine the squared Sharpe ratio for models with tradable factors. The most efficient asset pricing model should deliver the highest Sharpe ratio. We compute the Sharpe ratio as follows. Let μ denote the $K \times 1$ vector of mean values of the K pricing factors in a model, and Σ is the $K \times K$ variance-covariance matrix of pricing factors. For each three-factor model, we calculate the full-sample maximum Sharpe ratio (SR_{FULL}) as $SR_{FULL} = \sqrt{\hat{\mu}_F' \hat{\Sigma}_F \hat{\mu}_F}$, where $\hat{\mu}_F$ and $\hat{\Sigma}_F$ are the estimated mean vector and variance-covariance matrix of risk factors over the full sample period.

The last column of Table 1 reports the Sharpe ratio of value or profitability factor models. We observe that the profitability-based factor models produce higher Sharpe ratios than the value-based factor models. Among the profitability-based factor models, the factor constructed using EBIT/NOA (*ebit_bev*) produces the highest Sharpe ratio of 0.4305. Among the value-based factors, cash flow-to-price measures generate the highest Sharpe ratios with FCF/ME (*fcf_me*) and OCF/ME (*ocf_me*) generating Sharpe ratios of 0.3672 and 0.3653, respectively.

Our second methodology to select robust risk factors is the pairwise regression spanning test. A robust risk factor is the candidate factor that has incremental ability to price the cross-section of stock returns. It should span the risk premium of other candidate risk factors and remain not subsumed by other risk factors. Following Ball et al. (2016), we first regress the return of the focal factor returns on another factor returns in the same cluster. The focal factor is ‘subsumed’ by the other factor under the condition that the focal factor return becomes insignificant after adjusting for the other factor. Then, we conduct the exercise in reverse. The focal factor ‘spans’ the other

factor under the condition that the other factor return is no longer significant after adjusting for the focal factor.

Table 2 tabulates results of the spanning tests. Panel A reports the results of the seven profitability factors. The upper panel examines whether the focal factor is subsumed by the other six profitability factors. The first row [1] (*ebit_bev*) shows EBIT/NOA factor is not subsumed by any of the other six profitability factors: It continues to earn significantly positive returns after adjusting for all six profitability factors individually, indicating that no other profitability factor can account for the high average return of the EBIT/NOA-based factor. The last column shows that after adjusting for other profitability risk factors, EBIT/NOA exhibits an average abnormal return of 0.743% per month, which is the largest among the candidate factors.

The lower panel examines whether the focal factor spans the other six profitability factors. The first row [1] of the lower panel ($x=ebit_bev$) shows that EBIT/NOA spans all other profitability factors because none of the other six profitability factors remain significant at better than the 5% confidence level after regressing on EBIT/NOA. The last column shows that EBIT/NOA reduces the absolute alpha of the other six profitability factors to only 0.247% per month. This is the lowest absolute alpha among the candidate profitability factors.

Overall, EBIT/NOA (*ebit_bev*) is the most robust profitability factor and exhibits the unique ability to span and remain not subsumed by any of the other candidate factors. For comparison, the ‘weakest’ profitability factor is profit margin (*ebit_sale*). The seventh row [7] of the upper panel ($y=ebit_sale$) shows that it is subsumed by all other profitability factors. In the seventh row [7] of the lower panel ($y=ebit_sale$), profit margin-adjusted returns fail to span the other factors in the profitability cluster. Notably, no other candidate risk factor in the profitability cluster exhibits these two unique properties.

Panel B reports results for the value cluster. The upper panel examines whether the focal factor is subsumed by the 12 other value factors. The best performers are the cash flow-to-price measures in row [3] ($x=ocf_me$) and row [4] ($x=fcf_me$). Both measures are not subsumed by any of the other 12 candidate value factors and continue to earn large absolute alphas of 1.247% and 1.393%, respectively. The lower panel presents the corresponding spanning test. The best performer is in row [3] ($x=ocf_me$), which spans 11 of the 12 other value factors. The other cash flow-to-price measure in row [4] ($x=fcf_me$) underperforms in this test because it spans only 4 of the other 12 factors.

Overall, operating cash flow-to-price (ocf_me) is the most robust value factor. It is unique among the candidate factors in that it is not subsumed by any other factor and spans 11/12 other value factors.

Based on the results of the two tests, we select EBIT/NOA and OCF/ME to create the profitability and value factors, respectively. It is worth noting that our selection is statistical in that ex-ante we have no prior belief on which characteristics would produce the most robust factor in the Indonesian stock market. In other markets, book-to-market, earnings-to-price, and gross profitability are the most robust factors. We speculate that the success of EBIT/NOA and OCF/ME could relate to local accounting standards, listing regulations, and industry composition, although these mechanisms are not directly testable.

2.5. Creating a four-factor asset pricing model

We propose a four-factor asset pricing model that consists of the market, size, value (OCF/ME), and profitability (EBIT/NOA) factors. We use the 2-by-2-by-2 procedure following Fama and French (2015) to construct orthogonal SMB, HML, and RMW factors. Specifically, each month we sort individual stocks into 8 portfolios based on breakpoints from the median values

of firm size, value, and profitability ratios using stocks on the Main board. The size, value, and profitability factors (SMB, HML, and RMW) are the value-weighted average of returns on the four small-minus-big portfolios, four high-minus-low portfolios, and four robust-minus-weak portfolios in the next month.

3. Factor model performance tests

To assess the performance of the IDX4 model, we benchmark the IDX4 against the following seven models: (1) the single-factor CAPM, (2) a three-factor model including MKT, SMB, and HML (IDX3-size-value), and (3) a three-factor model including MKT, SMB, and RMW (IDX3-size-prof). We also create an IDX version of established factor models developed for the U.S. market: (4) the Fama and French (1993) three-factor model (FF3) which adds the size and value (based on book-to-market ratio) to the CAPM, (5) the Carhart (1997) four-factor model (CAR4) which augments the FF3 with the momentum factor (based on cumulative stock return in the previous 2 to 12 months), (6) the q-factor model of Hou, Xue, and Zhang (2015), and (7) the Fama and French (2015) five-factor model adding profitability (based on operating profitability) and investment (based on total asset growth rate) factors.

3.1. Cross-sectional tests: Digesting anomalies

A successful asset pricing model should ‘digest’ the cross-sectional return predictability of anomalies such that the associated portfolio return is insignificant after accounting for the factor model. We follow JKP (2023) to construct 149 LS portfolios, which are grouped into 13 clusters.

⁷ Each LS portfolio buys the most underpriced stocks (in the top quintile portfolio) and sells the most overpriced stocks (in the bottom quintile portfolio). For each LS portfolio with a significantly

⁷ JKP (2023) cover 152 anomalies. 3 out of the 152 anomalies are dropped due to lack of sufficient data in the Indonesian market.

positive average return, we regress LS portfolio return on a constant term and the risk factors of each of the eight models to be tested. The estimated regression intercepts are the alphas of long-short portfolios relative to each model. An insignificant and small alpha indicates that the model can capture the difference in expected returns between the two extreme portfolios sorted by the anomaly variable.

We observe significantly positive average returns in 32 of 149 LS portfolios during our sample period. Firm characteristics related to market valuation of equity and quality possess the strongest predictive power for future stock returns. Long-short portfolios based on 12 value-related variables and 9 quality-related variables earn significant average returns. Five variables related to the level and growth of firm profitability also significantly affect cross-sectional expected returns on IDX stocks. We do not observe significant momentum effects, short-term reversals, or low risk anomalies during our sample period.

Table 3 summarizes the performance of the eight candidate factor models. For each model, we report the number of LS portfolios with insignificant alphas (out of the 32 significant LS portfolios) and the means of absolute values of LS alphas relative to each model by all anomalies and by clusters of anomalies. We tabulate the full set of alphas of all LS portfolios and associated t -statistics to test the null hypothesis that a portfolio alpha is zero in Table IA3.

Our IDX4 model digests 29 of the 32 anomalies. After adjusting for the four factors, the average magnitude of LS portfolio alphas is 0.514%, which is the lowest among the eight factors examined. Only 3 of the 32 LS portfolios are unexplained by the IDX4: 1) research and development expense to sales ratio (*rd_sale*), 2) market capitalization (*me*), and 3) net issuance of equity to asset ratio (*eqnetis_at*). Grinblatt and Saxena (2018) document that many factors cannot span their basis portfolios. Hence it is a notable achievement that the IDX4 can span the OCF/ME

and EBIT/NOA basis portfolios. However, it cannot span the size-based portfolio. Similarly, the other candidate factor models also fail to price the size-based portfolio. A possible reason is that traditional weighting in the construction of the factor-mimicking portfolio is unlikely optimal.

It is worth assessing the incremental pricing ability of IDX4 relative to a three-factor version of the model. We find that the IDX3 (size+value) and IDX3 (size+profitability) models are not as successful as the IDX4 model. They explain the predictability of 20 of the 32 and 21 of the 32 anomalies, respectively. IDX3 (size+value) digests 9 out of 12 significant value anomalies and reduce the average alphas of value-based LS portfolios to 0.521%. For comparison, the CAPM produces average alphas of value-based LS portfolios of 1.161%. The model also successfully digests all LS portfolios based on accruals, investment, and seasonality. The IDX3 (size+profitability) model digests 3 out of 3 profitability anomalies. It also completely digests 9 out of 9 qualities anomalies. Similar to the IDX3 (size+value) model, the IDX3 (size+profitability) model successfully digests all LS portfolios based on accruals, investment, and seasonality, but it fails to explain many of the value anomalies. The noticeable improvement of the IDX4 model over the individual IDX3 models suggests that the four-factor model is necessary to explain the cross-section of stock returns on the IDX.

The other factor models underperform the IDX4. The single-factor CAPM only explains the predictability of 14 of the 32 anomalies. It performs best in explaining differences in returns of investment, low leverage, and seasonality anomalies. The other models explain 13 (FF3), 12 (CAR4), 20 (Q4), and 15 (FF5) of the 32 anomalies. It is worth noting that the FF3, CAR4, and FF5 are constructed using the book-to-market ratio because BM constructed factors perform the best in the U.S. setting. In other markets, valuations ratios such as EP work best empirically (Liu, Stambaugh, and Yuan, 2019). In our earlier analysis, we document that the CP-based factor works

best in the Indonesia setting. As such, it is perhaps not surprising that these models do not perform as well as IDX3 (size+value) constructed using CP. Considering that the Q4 model is off-the-shelf, it performs quite well in capturing 20 of the 32 anomalies.

Overall, our evidence indicates that the IDX4 model provides significant improvements over the CAPM in digesting cross-sectional stock anomalies on the IDX. Our local version factor models outperform U.S.-based existing models.

3.2. Time-series tests: GRS test

We evaluate the ability of the candidate factor models to explain the time-series variation in stock returns using the GRS test. Table 4 presents the results. We start by analyzing the time-series return of the 32 LS anomaly portfolios. Panel A reports the number of instances that the model passes the GRS test statistic with a p -value greater than 0.01, 0.05, and 0.10. Specifically, we test the null hypothesis that the alphas relative to each candidate model is equal to zero. The bottom row reports the average absolute value of alphas of all 32 LS portfolios. The IDX4 and IDX3 size+value models perform relatively better than the other candidate models in explaining average returns on equity portfolios in terms the number of instances the models pass the GRS test statistic. IDX4 passes the GRS test with p -values greater than 10% (1%) in 8 (4) out of the 32 anomaly portfolios. It produces the second lowest average absolute alphas. The average portfolio alpha of the 32 LS portfolios is 0.440% and 0.430% for IDX4 and IDX3 size+value, respectively. In comparison, the average absolute alpha is 0.675% for CAPM. FF3, CAR4, Q4, and FF5 perform better than the CAPM, but not as well as IDX4.

Panel B of Table 4 reports the same analysis for 5×5 portfolios sorted by size $\times$ value and size $\times$ profitability. Relatively, IDX4 is the strongest performer of the candidate models. IDX4 model has the lowest GRS test statistics for both the size $\times$ value and size $\times$ profitability portfolios

(3.811 and 6.299, respectively). The CAPM has the worst GRS test statistic of 9.735 and 9.454, respectively. We observe that FF3, CAR4, Q4, and FF5 perform better than the CAPM, but not as well as IDX4.

Panel C of Table 4 repeats the test using 11 industry portfolios.⁸ Again, we observe that the IDX4 and IDX3 size+value perform the best among the candidate models. However, the performance of the models is poor across all eight models.

Overall, the GRS test results suggest that the IDX4 model generally improves performance over the single-factor CAPM. Although the null hypothesis of jointly zero portfolio alphas is rejected in most cases, our findings are comparable to prior studies conducted on the U.S. market. It is well known that time-series variation in stock returns presents a greater challenge for factor models than cross-sectional variation.

3.3. Time-series tests: Out-of-sample tests

The GRS test evaluates the in-sample ability of a factor model to describe equity portfolio returns. To assess the viability of trading strategies, we need to examine the out-of-sample ability of a model to predict future return. For out-of-sample model fitness, we compute the following two out-of-sample R^2 as test statistics,

$$\text{Total } R^2 = 1 - \frac{\sum_{i,t} (r_{t+1}^i - \hat{\beta}_t^{i'} \hat{f}_{t+1})^2}{\sum_{i,t} (r_{t+1}^i)^2}, \quad (1)$$

$$\text{predictive } R^2 = 1 - \frac{\sum_{i,t} (r_{t+1}^i - \hat{\beta}_t^{i'} \hat{\lambda})^2}{\sum_{i,t} (r_{t+1}^i)^2}, \quad (2)$$

⁸ The IDX classifies industries into the following eleven sectors: Basic Materials Consumer Cyclical, Consumer Non-Cyclical, Energy, Financials, Healthcare, Industrials, Infrastructures, Properties & Real Estate, Technology, and Transportation & Logistic.

where i denotes an asset and t denotes a month. r_{t+1}^i is the excess monthly asset return in the next month. $\hat{\beta}_t^i$ is a vector of beta exposures to risk factors in a model of assets estimated using information up to month t . $\hat{f}_{t+1}$ is a vector of factor return on the next month. $\hat{\lambda}$ is a vector of estimated risk premiums on factors in a model, which is the average value of factor returns from the first month up to month t . The two R^2 statistics measure the proportional of total variation in asset returns that can be explained by risk exposures and realizations of risk factors and by factor loadings and estimated factor risk premiums. We consider four sets of test assets: (1) 25 portfolios sorted by size and value, (2) 25 portfolios sorted by size and profitability, (3) 10 industry portfolios, and (4) quintile portfolios sorted by the 32 significant stock return predictors.

Table 5 reports the out-of-sample total and predictive R^2 for the eight candidate models. Panel A shows that IDX4 generates the highest total R^2 in explaining total variation in returns on 25 size $\times$ value and 25 size $\times$ profitability portfolios out-of-sample, which are 10% higher than the total R^2 from the CAPM. The CAPM provides the worst model fitness in explaining equity portfolios sorted by size and value and size and profitability variables with a total R^2 of 39.16% and 40.05%, lowest among all models. IDX4 also generates the highest total R^2 for industry portfolio in column 3 and highest average total R^2 of the 32 LS characteristics-sorted portfolios in column 4 of 69.58% compared to 65.29% from the CAPM.

Using predictive out-of-sample R^2 , IDX4 remains a strong performer. It exhibits predictive out-of-sample R^2 of 2.11% and 1.81%, which is second best to IDX3 size+value with 2.18% and 2.01% for the 25 size $\times$ value and 25 size $\times$ profitability portfolios. Although IDX4 has middle-of-pack performance for industry portfolios, it is the best performer with the highest average predictive R^2 of the 32 LS characteristics-sorted portfolios in column 4.

In sum, the results of out-of-sample tests suggest that the IDX4 is the stronger performer of the candidate factor models. It is consistent with the results from the spanning tests and GRS tests.

3.4. Model-implied expected returns versus actual returns

Another methodology to evaluate factor model performance is to compare model-implied expected returns to actual average returns. For each testing portfolio and each model, we first compute model implied expected returns as follows. For a portfolio i in a month t , excess returns on the portfolios are regressed on risk factors of a model in a rolling window of 24 to 60 months (as available) to estimate the portfolio's conditional factor loadings, $\hat{\beta}_t^i$. The expected excess return on a portfolio i for the next month $t + 1$ formed in month t is $E_t(r_{t+1}^i) = \hat{\beta}_t^{i'} \hat{\lambda}_t$, where the estimated factor premium $\hat{\lambda}_t$ is average returns on factors during the rolling window.

Figure 1 presents scatterplots of the average model-implied expected return against the average realized returns on the 25 portfolios sorted by size and value, 25 portfolios sorted by size and profitability, and industry portfolios. A well performing model will produce dots that lie along the 45-degree line on the scatterplot.

Panel A displays the scatterplots for each model using 25 portfolios sorted by size and value. The IDX4 model fits closest to the 45-degree line among the candidate models. It exhibits the steepest slope, indicating that it provides the closest fit of expected to actual returns. It is successful in producing low expected returns on large growth stocks and high expected returns on value stocks with intermediate size, but it struggles with small stocks with the highest market valuation ratio. The IDX4 model is a better fit than the IDX3 size+value and IDX3 size+profitability models. The IDX3 models struggle to explain the high average return on small stocks with high value or the low average return on large stocks with low value. The CAPM

produces the worse fit, with a flat and slightly negative relation between implied expected return and actual average return. Regardless of size and value, the CAPM produces similar expected returns on all portfolios.

We observe similar patterns in panel B using 25 portfolios sorted by size and profitability. Again, the IDX4 model fits closest to the 45-degree line among the candidate models and exhibits the steepest slope. It produces low expected returns on large growth stocks and high expected returns on value stocks with intermediate size, but struggles with small stocks with the highest market valuation ratio. We also find that the IDX4 model is a better fit than the IDX3 size+value and IDX3 size+profitability models. The IDX3 models struggle to explain the high average return on small stocks with high value or the low average return on large stocks with low value. The CAPM produces the worse fit, with a flat and slightly negative relation between implied expected return and actual average return. Regardless of size and value, the CAPM produces similar expected returns on all portfolios.

The 11 industry portfolios provide a challenge for all factor models. All 8 factor models produce flat or even negative slopes. Again, the results are consistent with prior studies that find that factors models are unable to price industry portfolios.

Overall, the differences in average returns of stocks with different firm sizes, market valuation of equity, and firm profitability can be largely explained by their different exposures to size, value, and profitability factors. However, differences in average industry portfolio returns remain unexplained by the IDX4 model.

4. Additional tests

In this section, we use the unique features of the IDX market to conduct additional tests.

4.1. Sharia compliant stocks

Our first test examines the demand for Islamic financial assets in Indonesia, where Muslims accounts for more than 80% of the population. To cater to the investment needs of these investors, the IDX classifies stocks into two groups: (1) Sharia-compliant stocks are issued by firms operating in compliance with Islamic principles. (2) Non-Sharia-complaint stocks are issued by companies of whose operations are not in accordance with Islamic principles. Firms adhering to the Islamic laws exhibit different dividend payout (Ben-Nasr and Ghouma (2022)) and risk management policies (Majdoub and Sassi (2017)). Islamic equity markets can also mitigate illicit activities related to Bitcoin (Karim et al. (2023)). We expect that the preferences for Sharia stocks could create different pricing mechanisms for the two groups of stocks.

Table 6 reports the return predictability of LS characteristics-based portfolios by separately conducting quantile portfolio sorts for Sharia and non-Sharia stocks. Panel A reports that among Sharia stocks, there are 15 significant anomaly returns across 6 clusters. 5 of the 6 clusters overlap with the full-sample set of clusters with the low-risk cluster as the lone exception. IDX4 explains the most LS portfolio among the 8 candidate models, capturing 13 of the 15 significant long-short portfolios. Among Sharia stocks, IDX4 cannot account for the size anomaly and the change in sales-to-inventory ratio (*dsale_dinv*). It performs as well as or better than the other factor models in all 6 of the anomaly clusters. However, in terms of minimizing absolute alpha of the 15 anomalies, the performance is only middle of the pack.

Panel B reports 33 significant anomaly returns among non-Sharia stocks across 10 clusters. There are more anomalies among non-Sharia stocks relative to Sharia stocks (33 vs. 15). 8 of the 10 clusters overlap with the full-sample set of clusters. Debt issuance and momentum are significant anomalies that occur among non-Sharia stocks, but not in the full sample. The

significant momentum measures are formulated using the past three-month return and residual cumulative return in the past one year.

IDX4 continues to outperform other models in explaining anomalies within the non-Sharia subsample. The IDX4 model explains 31 of the 33 significant long-short strategies. IDX4 cannot account for the size anomaly and 3-month momentum anomaly (*ret_3_1*) among non-Sharia compliant stocks. It reduces model-adjusted LS alphas to 0.636%, the lowest among all models and performs as well as or better than the other factor models in all 10 of the anomaly clusters.

4.2. Main Board listings

IDX has three classifications of stocks depending on trading history, firm profitability, and market liquidity. Stocks listed on the Main Board account for 55% of all listed stocks and consist of large companies with long trading history and active stock trading. The remaining 45% stocks that have not yet met the requirements of Main Board are listed on the Development or Acceleration Board.

We separately assess the return predictability associated with 149 firm characteristics for stocks listed and not listed on the Main Board. For stocks listed on the Main Board, 29 firm characteristics exhibit significant anomaly returns across 9 anomaly clusters. The anomaly clusters mostly (8/9) overlap with the full sample analysis (Table 3). For non-Main Board stocks, 17 firm characteristics exhibit significant anomaly returns across 4 anomaly clusters. All four clusters overlap with the clusters in the full sample (i.e., profit growth, profitability, size, value).

Table 7 reports the results. Panel A shows that for stocks listed on the Main Board, IDX4 produces the lowest average magnitude of model-adjusted return spreads between extreme portfolios with very different firm characteristics. IDX4 digests 25 of the 29 anomalies. After adjusting for the four factors, the average magnitude of LS portfolio alphas is 0.728%, which is

the second lowest among the eight factors examined. Only 4 of the 29 LS portfolios are unexplained by the IDX4: 1) Change in net operating assets (*noa_gr1a*), 2) Change in sales to inventory ratio (*dsale_dinv*), 3) Gross profits-to-assets ratio (*gp_at*), and 4) size (*me*). It performs as well as or better than the other factor models in all but one anomaly cluster (Quality).

Panel B reports 17 significant anomaly returns among non-Main Board stocks. This finding indicates that there are more anomalies among Main Board stocks relative to non-Main Board stocks (29 vs. 17). 4 of the 4 clusters overlap with the full-sample set of clusters. For stocks listed or not listed on the Main Board, the IDX4 produces the lowest average magnitude of model-adjusted return spreads between extreme portfolios of 0.709. It explains 16 of the 17 anomalies, with size remaining the lone exception.

In sum, our IDX4 model has the best overall performance in digesting anomalies regardless of the listing boards.

5. Conclusion

We propose a four-factor model to explain stock returns in the Indonesian stock market. The model contains the market-, size-, value-, and profitability-factors constructed from equity portfolios sorted by market capitalization, *OCF/ME*, and *EBIT/NOA*. We select the latter two factors because they are the best performing factors from factor-spanning tests, absolute alpha tests, GRS tests, and out-of-sample R^2 tests.

The IDX4 model is the best performer among the eight candidate models in digesting return anomalies. It can explain the average returns of 29 out of 32 significant long-short characteristics-based portfolios. The model also generates the best full-sample and out-of-sample model fit for returns on characteristics-sorted portfolios. Expected returns on portfolios implied by the IDX4 model are closest to average realized returns on 25 portfolios sorted by firm size and value and

profitability variables. We also find that the IDX4 model performs best within the following subsamples: 1) Sharia stocks, 2) non-Sharia stocks, 3) Mainboard listing, and 4) non-Mainboard listings.

We acknowledge two limitations of the IDX4 model. First, the model struggles to explain average returns on industry portfolios. Second, recent studies propose the use of machine learning (ML) and dimension reduction techniques to identify the most relevant factors from high-dimensional and highly correlated factors. We do not use an ML-based approach because it requires a relatively long trading history for reliable inference. With a longer time-series of returns, we expect future researchers to employ ML techniques to build on the findings in this study.

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Table 1. Average returns and Sharpe ratios of candidate risk factors

The table reports the average monthly returns in percentage of the market, size, value, and profitability factors in the IDX market. The t -statistic tests the null hypothesis of zero average factor returns. *mkt* is the market excess return defined as the difference between the monthly return on the value-weighted equity portfolio of all IDX stocks and the monthly risk-free rate. *me* is the size premium, which is the difference between average return of bottom and top quintiles sorted by market capitalization. We use a 2-by-3 sorting procedure to construct value and profitability factors using measures of valuation and firm profitability. Each value factor is the average return on the two long-short portfolios that buys the value portfolios sells the growth portfolios (HML). Each profitability factor is the average returns of the two long-short portfolios that buys the robust portfolios and sells the weak portfolios (RMW). The last column reports the Sharpe ratio of the three-factor model that includes the market (MKT), size (SMB), and one of the value or profitability factors constructed based on various value and profitability measurements using the 2-by-3 procedure of Fama and French (1993). The sample period is January 1990 to June 2023.

Cluster	Characteristic	Acronym	Mean	t -stat	SMB		Sharpe ratio 3-factor model
					Mean	t -stat	
Market	Market excess return	<i>mkt</i>	1.406	(3.48)			-
Size	Market equity	<i>me</i>			1.629	(3.01)	-
Value	Net equity issuance	<i>eqnetis_at</i>	2.166	(2.21)	1.548	(2.01)	0.2937
	EBITDA/EV	<i>ebitda_mev</i>	1.934	(4.78)	0.148	(0.43)	0.3059
	Operating cash flow-to-price	<i>ocf_me</i>	1.764	(6.39)	0.397	(1.06)	0.3653
	Free cash flow-to-price	<i>fcf_me</i>	1.609	(5.41)	0.462	(1.25)	0.3672
	Net total issuance	<i>netis_at</i>	1.377	(3.91)	0.525	(1.22)	0.2997
	Sales-to-price	<i>sale_me</i>	1.350	(3.38)	0.075	(0.20)	0.2464
	Total assets-to-price	<i>at_me</i>	1.310	(2.53)	0.337	(0.97)	0.2461
	Equity duration	<i>eq_dur</i>	1.285	(3.59)	-0.048	(0.14)	0.2846
	Book-to-market equity	<i>be_me</i>	1.285	(2.16)	0.087	(0.22)	0.2447
	Intrinsic value-to-price	<i>ival_me</i>	1.198	(3.27)	0.281	(0.76)	0.2870
	Net payout yield	<i>eqnpoy_me</i>	1.133	(2.75)	0.510	(1.14)	0.2586
	Earnings-to-price	<i>ni_me</i>	0.835	(2.10)	0.379	(1.04)	0.2860
	Dividend yield	<i>div12m_me</i>	0.908	(2.71)	0.060	(0.20)	0.2446
Profitability	EBIT/NOA	<i>ebit_bev</i>	1.232	(3.73)	0.916	(2.44)	0.4305
	Operating cash flow-to-assets	<i>ocf_at</i>	1.084	(3.92)	0.628	(1.55)	0.3515
	Operating profits-to-lagged book equity	<i>ope_bell</i>	1.006	(3.33)	0.614	(1.43)	0.3298
	Return on equity (quarterly)	<i>niq_be</i>	0.981	(2.61)	0.714	(2.40)	0.4047
	Operating profits-to-book equity	<i>ope_be</i>	0.835	(2.60)	0.874	(2.19)	0.3419
	Return on equity (annual)	<i>ni_be</i>	0.786	(2.58)	0.873	(2.24)	0.3265
	Profit margin	<i>ebit_sale</i>	0.768	(2.59)	0.751	(1.91)	0.3017

Table 2. Spanning tests on value and profitability factors

We regress monthly returns from one factor (f_Y) on a constant term and monthly returns from a different factor within the same cluster (f_X). The intercept estimate from the regression is the alpha of f_Y after adjusting for the other factor f_X , α_X^Y . The table shows alphas of factors relative to other factors for the profitability cluster (in Panel A) and the value cluster (in Panel B). Superscript ¹, ⁵, and ¹⁰ indicates significant factor alphas at 1%, 5%, and 10% significance level. The last two columns report the number of insignificant α_X^Y and the mean absolute value of α_X^Y (in the upper section) and the number of insignificant α_Y^X and the mean absolute value of α_Y^X (in the lower section). Factor alphas are reported in percentage. The sample period is January 1990 to June 2023.

Panel A. Profitability factors (market + size + profitability factor)

		X variable							# $t(\alpha_X^Y) < 1.96$	$ \overline{\alpha_X^Y} $
		[1]	[2]	[3]	[4]	[5]	[6]	[7]		
Y variable	[1] <i>ebit_bev</i>		0.740 ¹	0.788 ¹	0.787 ¹	0.667 ¹	0.611 ¹	0.862 ¹	0	0.743
	[2] <i>ocf_at</i>	0.356 ¹⁰		0.449 ⁵	0.576 ¹	0.346 ⁵	0.456 ¹	0.762 ¹	1	0.491
	[3] <i>ope_bell</i>	0.280	0.320 ¹⁰		0.493 ⁵	0.061	0.358 ¹⁰	0.717 ¹	4	0.372
	[4] <i>niq_be</i>	0.350	0.705 ⁵	0.480		0.386	0.509 ⁵	0.878 ¹	3	0.551
	[5] <i>ope_be</i>	0.231	0.309 ¹⁰	0.159	0.501 ⁵		0.382 ⁵	0.774 ¹	3	0.393
	[6] <i>ni_be</i>	0.058	0.352	0.388	0.487 ¹⁰	0.304		0.658 ¹	5	0.375
	[7] <i>ebit_sale</i>	-0.209	0.240	0.344	0.287	0.278	0.158		6	0.253
		Y variable							# $t(\alpha_Y^X) < 1.96$	$ \overline{\alpha_Y^X} $
		[1]	[2]	[3]	[4]	[5]	[6]	[7]		
X variable	[1] <i>ebit_bev</i>		0.356 ¹⁰	0.280	0.350	0.231	0.058	-0.209	6	0.247
	[2] <i>ocf_at</i>	0.740 ¹		0.320 ¹⁰	0.705 ⁵	0.3091	0.352	0.240	4	0.444
	[3] <i>ope_bell</i>	0.788 ¹	0.449 ⁵		0.480	0.159	0.388	0.344	4	0.435
	[4] <i>niq_be</i>	0.787 ¹	0.576 ¹	0.493 ⁵		0.5015	0.487 ¹⁰	0.287	3	0.522
	[5] <i>ope_be</i>	0.667 ¹	0.346 ⁵	0.061	0.386		0.304	0.278	4	0.34
	[6] <i>ni_be</i>	0.611 ¹	0.456 ¹	0.358 ¹	0.509 ⁵	0.3825		0.158	2	0.412
	[7] <i>ebit_sale</i>	0.862 ¹	0.762 ¹	0.717 ¹	0.878 ¹	0.7741	0.658 ¹		1	0.775

Table 2. Spanning tests on value and profitability factors (continued)

Panel B. Value factors (market + size + value factor)

		X Variable													# $t(\alpha_X^Y) < 1.96$	$ \overline{\alpha_X^Y} $
		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]		
Y- variable	[1]eqnetis_at		1.251 ¹	1.004 ⁵	0.787 ¹⁰	0.857 ¹⁰	1.158 ¹	1.010 ⁵	1.119 ¹	1.103 ¹	1.287 ¹	0.878 ⁵	1.258 ¹	1.294 ¹	2	1.084
	[2]ebitda_mev	1.433 ¹		0.246	1.312 ¹	1.363 ¹	0.947 ¹	0.881 ¹	0.658 ¹	1.037 ¹	0.930 ¹	1.446 ¹	0.535 ⁵	1.203 ¹	1	0.999
	[3]ocf_me	1.584 ¹	0.767 ¹		1.001 ¹	1.362 ¹	1.300 ¹	1.316 ¹	1.097 ¹	1.417 ¹	1.317 ¹	1.531 ¹	0.869 ¹	1.403 ¹	0	1.247
	[4]fcf_me	1.439 ¹	1.424 ¹	0.841 ¹		1.172 ¹	1.572 ¹	1.480 ¹	1.514 ¹	1.602 ¹	1.494 ¹	1.399 ¹	1.312 ¹	1.462 ¹	0	1.393
	[5]netis_at	0.925 ¹	1.161 ¹	0.644 ¹⁰	0.512 ¹⁰		1.134 ¹	1.236 ¹	1.219 ¹	1.290 ¹	1.243 ¹	1.200 ¹	1.177 ¹	1.212 ¹	2	1.079
	[6]sale_me	0.939 ¹	0.154	0.207	1.010 ¹	0.771 ¹		0.487 ¹⁰	0.232	0.399 ¹⁰	0.588 ⁵	0.901 ¹	0.398	0.847 ¹	6	0.578
	[7]at_me	1.050 ¹	-0.004	0.206	0.855 ¹	0.892 ¹	0.478 ¹⁰		0.258	0.375 ¹⁰	0.588 ⁵	0.926 ¹	0.378	0.903 ¹	6	0.576
	[8]eq_dur	1.177 ¹	0.278	0.334	1.267 ¹	1.226 ¹	0.715 ¹	0.761 ¹		0.724 ¹	0.811 ¹	1.353 ¹	0.303	1.108 ¹	3	0.838
	[9]be_me	1.149 ¹	0.132	0.360	1.218 ¹	0.996 ¹	0.282	0.274	0.062		0.680 ⁵	1.037 ¹	0.454	0.981 ¹	6	0.635
	[10]ival_me	1.340 ¹	0.356	0.477	1.108 ¹	1.120 ¹	0.832 ¹	0.830 ¹	0.584 ¹⁰	0.930 ¹		1.161 ¹	0.454	0.831 ¹	4	0.835
	[11]eqnpo_me	0.875 ¹⁰	0.982 ¹⁰	0.421	0.403	0.634	0.959 ⁵	0.912 ¹⁰	1.154 ¹	0.996 ⁵	0.908 ¹⁰		0.990 ¹⁰	0.830 ¹⁰	9	0.839
	[12]ni_me	1.383 ¹	0.471 ⁵	0.319	1.124 ¹	1.331 ¹	1.058 ¹	1.057 ¹	0.618 ¹	1.140 ¹	0.940 ¹	1.400 ¹		1.158 ¹	1	1.000
	[13]div12m_me	0.940 ¹	0.468	0.311	0.781 ⁵	0.842 ¹	0.850 ¹	0.881 ¹	0.668 ⁵	0.917 ¹	0.517 ¹⁰	0.881 ¹	0.469		4	0.710
		Y Variable													# $t(\alpha_Y^X) < 1.96$	$ \overline{\alpha_Y^X} $
		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]		
X variable	[1]eqnetis_at		-1.433	-1.584	-1.439	0.925 ¹	-0.939	-1.050	1.177 ¹	-1.149	-1.340	-0.875	-1.383	-0.940	10	1.186
	[2]ebitda_mev	-1.251		0.767 ¹	1.424 ¹	-1.161	0.154	-0.004	-0.278	0.132	0.356	0.982 ¹⁰	0.471 ⁵	0.468	9	0.621
	[3]ocf_me	-1.004	0.246		0.841 ¹	-0.644	0.207	0.206	-0.334	0.360	0.477	0.421	0.319	0.311	11	0.448
	[4]fcf_me	-0.787	1.312 ¹	1.001 ¹		-0.512	1.010 ¹	0.855 ¹	-1.267	1.218 ¹	1.108 ¹	0.403	1.124 ¹	0.781 ⁵	4	0.948
	[5]netis_at	0.857 ¹⁰	-1.363	-1.362	-1.172		-0.771	-0.892	1.226 ¹	-0.996	-1.120	-0.634	-1.331	-0.842	11	1.047
	[6]sale_me	-1.158	0.947 ¹	1.300 ¹	1.572 ¹	-1.134		0.478 ¹⁰	-0.715	0.282	0.832 ¹	0.959 ⁵	1.058 ¹	0.850 ¹	5	0.940
	[7]at_me	-1.010	0.881 ¹	1.316 ¹	1.480 ¹	-1.236	0.487 ¹⁰		-0.761	0.274	0.830 ¹	0.912 ¹⁰	1.057 ¹	0.881 ¹	6	0.927
	[8]eq_dur	1.119 ¹	-0.658	-1.097	-1.514	1.219 ¹	-0.232	-0.258		-0.062	-0.584	-1.154	-0.618	-0.668	10	0.765
	[9]be_me	-1.103	1.037 ¹	1.417 ¹	1.602 ¹	-1.290	0.399 ¹⁰	0.375 ¹⁰	-0.724		0.930 ¹	0.996 ⁵	1.140 ¹	0.917 ¹	5	0.994
	[10]ival_me	-1.287	0.930 ¹	1.317 ¹	1.494 ¹	-1.243	0.588 ⁵	0.588 ⁵	-0.811	0.680 ⁵		0.908 ¹⁰	0.940 ¹	0.517 ¹⁰	5	0.942
	[11]eqnpo_me	-0.878	1.446 ¹	1.531 ¹	1.399 ¹	-1.200	0.901 ¹	0.926 ¹	-1.353	1.037 ¹	1.161 ¹		1.400 ¹	0.881 ¹	3	1.176
	[12]ni_me	-1.258	0.535 ⁵	0.869 ¹	1.312 ¹	-1.177	0.398	0.378	-0.303	0.454	0.454	0.990 ¹⁰		0.469	9	0.716
	[13]div12m_me	-1.294	1.203 ¹	1.403 ¹	1.462 ¹	-1.212	0.847 ¹	0.903 ¹	-1.108	0.981 ¹	0.831 ¹	0.830 ¹⁰	1.158 ¹		4	1.103

Table 3. Digesting anomalies

This table reports the number of insignificant alphas $\# t(\alpha_{LS}) < 1.96$ and the average absolute alphas $|\overline{\alpha_{LS}}|$ in percentage of 32 characteristic-based anomaly portfolios for 8 candidate factor models. To form the long-short (LS) characteristics-based anomaly portfolio, we sort individual stocks into quantile portfolios by firm characteristics each month. The LS portfolio buys stocks with the highest return and sells stocks with the lowest return. The upper panel reports the results using all 32 LS portfolios, and the lower panel reports the results by anomaly clusters. The 32 LS portfolios are chosen from 149 candidate portfolios in JKP (2023). Details on the construction of the 32 LS portfolios are available in the Internet Appendix. We estimate full-sample time series regressions with LS portfolio returns as the dependent variables on a constant term and risk factors of a model to estimate LS alpha relative to the model. The sample period is January 1990 to June 2023.

All 32 LS portfolios

Models	$\# t(\alpha_{LS}) < 1.96$ (out of 32)	$ \overline{\alpha_{LS}} $
IDX4	29	0.514
IDX3 (size+value)	20	0.668
IDX3 (size+prof)	21	0.735
CAPM	14	0.917
FF3	13	0.976
CAR4	12	0.941
Q4	15	0.768
FF5	20	0.810

LS portfolios by clusters

$\# t(\alpha_{LS}) < 1.96$	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Accruals (1)	1	1	1	0	0	1	1	0
Investment (1)	1	1	1	1	1	0	1	1
Low Leverage (1)	0	0	0	1	1	0	1	1
Profit Growth (2)	2	0	1	1	0	1	1	0
Profitability (3)	3	2	3	2	0	0	2	0
Quality (9)	9	5	9	4	3	3	9	4
Seasonality (2)	2	2	2	2	2	2	2	2
Size (1)	0	0	0	0	0	0	0	0
Value (12)	11	9	4	3	6	5	3	7
Total (out of 32)	29	20	21	14	13	12	20	15

$ \overline{\alpha_{LS}} $	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Accruals	0.628	0.457	0.693	1.019	1.103	1.013	0.698	0.939
Investment	0.121	0.040	0.642	0.210	0.178	0.364	0.247	0.082
Low Leverage	1.873	1.874	2.124	1.198	1.225	1.604	1.811	1.611
Profit Growth	0.806	1.025	0.967	0.867	0.895	0.948	0.837	1.002
Profitability	0.363	0.783	0.083	1.025	1.455	1.309	0.605	0.957
Quality	0.400	0.773	0.247	0.776	1.059	0.937	0.490	0.834
Seasonality	0.341	0.451	0.167	0.137	0.219	0.219	0.201	0.108
Size	0.599	0.479	1.196	0.925	0.509	0.565	0.651	0.500
Value	0.521	0.521	1.177	1.161	1.009	0.989	1.183	0.736

Table 4. Time-series GRS Test

The table reports the results from GRS (1989) tests that evaluate the time-series performance of each candidate factor model. We analyze three sets of test assets: 32 long-short (LS) characteristics-sorted portfolios (Panel A), 25 portfolios sorted by size $\times$ value or size $\times$ profitability (Panel B), and 11 industry portfolios (Panel C). Panel A reports the # of instances that the model passes the GRS test with a p -value greater than 0.01, 0.05, or 0.10. The bottom row reports the average absolute value of alphas of all 32 LS portfolios. Panel B and panel C report the GRS test statistic and the corresponding p -values that tests the null hypothesis that the portfolios alphas relative to the factor model are jointly equal to zero. The sample period is January 1990 to June 2023.

Panel A. Summary of GRS test on 32 characteristics-sorted portfolios

	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
#($p \geq 0.01$)	8	5	0	0	3	3	3	1
#($p \geq 0.05$)	4	5	0	0	2	1	2	0
#($p \geq 0.1$)	4	2	0	0	0	1	1	0
$ \alpha $	0.44	0.43	0.782	0.675	0.549	0.575	0.617	0.545

Panel B. 25 Portfolios sorted by size $\times$ value or size $\times$ profitability

		IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
size $\times$ value	GRS Stat	3.811	4.166	8.622	9.735	6.014	5.668	7.396	8.830
	p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	0.340	0.458	1.202	1.010	0.628	0.653	0.627	0.571
size $\times$ profitability	GRS Stat	6.299	7.240	7.585	9.454	8.686	8.244	7.291	9.176
	p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	0.667	0.778	1.341	1.1492	0.849	0.858	0.788	0.786

Panel C. Industry portfolios

	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
GRS Stat	6.787	5.838	13.919	10.837	7.520	8.451	29.256	14.737
p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$ \alpha $	0.584	0.454	0.944	0.7394	0.573	0.645	0.744	0.536

Table 5: Out-of-sample R^2

The table reports out-of-sample predictive and total R^2 , which is the proportion of total variation in returns on four sets of test assets explained by the product of factor loadings and average factor returns and product of risk exposures and realizations of risk factors. Risk exposures and average risk premiums are estimated out-of-sample. R^2 are reported in percentage. The sample period is January 1990 to June 2023.

Panel A. Total R^2

	25 size $\times$ value	25 size $\times$ profitability	Industry portfolios	Average of 32 firm characteristic portfolios
IDX4	49.12	50.24	42.24	69.58
IDX3 size+value	44.71	43.64	41.21	67.91
IDX3 size+profitability	47.26	47.56	41.90	67.63
CAPM	39.16	40.05	41.42	65.29
FF3	45.92	47.00	40.86	68.76
CAR4	44.57	45.53	39.09	68.27
Q4	42.24	43.04	39.22	67.47
FF5	41.21	43.16	37.50	67.83

Panel B. Predictive R^2

	25 size $\times$ value	25 size $\times$ profitability	Industry portfolios	Average of 32 firm characteristic portfolios
IDX4	2.11%	1.81%	0.45%	2.10%
IDX3 size+value	0.91%	0.48%	0.10%	1.36%
IDX3 size+profitability	2.18%	2.01%	1.01%	1.71%
CAPM	1.75%	1.68%	0.92%	1.63%
FF3	1.19%	1.03%	0.31%	-0.91%
CAR4	1.17%	0.85%	0.24%	-1.02%
Q4	1.55%	1.00%	-0.11%	-0.72%
FF5	1.42%	1.33%	0.09%	-0.92%

Table 6: Digesting anomalies: Sharia versus non-Sharia compliant stocks

This table separates the sample into Sharia and non-Sharia compliant stocks and reports the number of insignificant alphas $\# t(\alpha_{LS}) < 1.96$ and the average absolute alphas $|\overline{\alpha_{LS}}|$ in percentage of characteristic-based anomaly portfolios for 8 candidate factor models. To form the long-short (LS) characteristics-based anomaly portfolio, we sort individual stocks into quantile portfolios by firm characteristics each month. The LS portfolio buys stocks with the highest return and sells stocks with the lowest return. The upper panel reports the results using all LS portfolios, and the lower panel reports the results by anomaly clusters. The LS portfolios are chosen based on statistical significance from 149 candidate portfolios in JKP (2023). Details on the construction of the LS portfolios are available in the Internet Appendix. We estimate full-sample time series regressions with LS portfolio returns as the dependent variables on a constant term and risk factors of a model to estimate LS alpha relative to the model. The sample period is January 1990 to June 2023.

Panel A: Sharia compliant stocks**All 15 LS portfolios**

Models	$\# t(\alpha_{LS}) < 1.96$ out of 15	$ \overline{\alpha_{LS}} $
IDX4	13	0.899
IDX3 size+value	10	0.742
IDX3 size+profitability	8	1.005
CAPM	8	0.899
FF3	9	0.827
CAR4	10	0.787
Q4	8	0.877
FF5	10	1.027

LS portfolios by cluster

$\# t(\alpha_{LS}) < 1.96$	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Low Risk (1)	1	0	1	1	1	1	1	1
Profit Growth (2)	1	0	1	1	1	1	1	1
Quality (1)	1	1	1	1	0	1	1	1
Seasonality (1)	1	1	1	1	1	1	1	1
Size (1)	0	0	0	0	0	0	0	0
Value (9)	9	8	4	4	6	6	4	6
Total (out of 15)	13	10	8	8	9	10	8	10

$ \overline{\alpha_{LS}} $	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Low Risk	0.422	0.999	0.354	1.001	0.820	0.503	0.525	0.853
Profit growth	0.935	1.135	0.753	0.877	0.893	0.820	0.695	1.010
Quality	0.214	0.759	0.170	0.787	1.115	0.907	0.218	0.829
Seasonality	0.229	0.097	0.275	0.384	0.455	0.423	0.124	0.377
Size	0.736	0.870	1.368	1.175	0.652	0.677	1.174	0.675
Value	1.113	0.682	1.268	0.932	0.842	0.851	1.081	1.184

Table 6: Digesting anomalies: Sharia versus non-Sharia compliant stocks (continued)**Panel B: Non-Sharia compliant stocks****All 33 LS portfolios**

Models	# $t(\alpha_{LS}) < 1.96$ out of 33	$ \alpha_{LS} $
IDX4	31	0.636
IDX3 size+ value	31	0.671
IDX3 size+prof	21	0.979
CAPM	27	1.070
IDX4	31	0.636
FF3	20	1.106
CAR4	19	1.097
Q4	30	0.825
FF5	24	0.939

LS portfolios by cluster

# $t(\alpha_{LS}) < 1.96$	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Accruals (1)	1	1	1	1	1	0	1	1
Debt Issuance (2)	2	2	2	2	2	2	2	2
Investment (3)	3	3	3	3	3	3	3	3
Momentum (2)	1	1	1	1	1	1	1	1
Profit Growth (2)	2	2	1	2	2	2	2	2
Profitability (5)	5	5	3	5	3	3	5	4
Quality (5)	5	5	5	5	2	2	5	2
Seasonality (1)	1	1	1	1	1	1	1	1
Size (1)	0	0	0	0	0	0	0	0
Value (11)	11	11	4	7	5	5	10	8
Total (out of 33)	31	31	21	27	20	17	30	24

$ \alpha_{LS} $	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Accruals	0.225	0.082	0.822	0.775	0.939	1.310	0.890	0.747
Debt issuance	0.699	0.242	0.172	0.240	0.104	0.255	0.064	0.329
Investment	0.364	0.204	0.445	0.455	0.497	0.315	0.514	0.311
Momentum	1.383	1.448	1.227	1.568	1.643	1.255	1.478	1.620
Profit growth	0.728	0.712	0.783	0.449	0.414	0.518	0.544	0.515
Profitability	0.623	0.709	0.998	0.844	0.926	1.012	0.793	0.590
Quality	0.411	0.352	0.302	0.905	1.356	1.213	0.371	1.096
Seasonality	0.230	0.471	0.088	0.895	1.169	1.025	0.045	1.149
Size	2.405	2.521	3.021	2.856	2.391	2.463	2.440	2.318
Value	0.569	0.759	1.471	1.469	1.345	1.390	1.121	1.134

Table 7: Digesting anomalies: Mainboard versus non-Mainboard listings

This table separates the sample into Mainboard and non-Mainboard lists and reports the number of insignificant alphas $\# t(\alpha_{LS}) < 1.96$ and the average absolute alphas $|\overline{\alpha_{LS}}|$ in percentage of characteristic-based anomaly portfolios for 8 candidate factor models. To form the long-short (LS) characteristics-based anomaly portfolio, we sort individual stocks into quantile portfolios by firm characteristics each month. The LS portfolio buys stocks with the highest return and sells stocks with the lowest return. The upper panel reports the results using all LS portfolios, and the lower panel reports the results by anomaly clusters. The LS portfolios are chosen based on statistical significance from 149 candidate portfolios in JKP (2023). Details on the construction of the LS portfolios are available in the Internet Appendix. We estimate full-sample time series regressions with LS portfolio returns as the dependent variables on a constant term and risk factors of a model to estimate LS alpha relative to the model. The sample period is January 1990 to June 2023.

Panel A: Mainboard listed stocks**All 29 LS portfolios**

Models	$\# t(\alpha_{LS}) < 1.96$ out of 29	$ \overline{\alpha_{LS}} $
IDX4	25	0.728
IDX3 size+value	21	0.724
IDX3 size+profitability	18	0.798
CAPM	14	0.883
FF3	14	0.888
CAR4	16	0.827
Q4	20	0.733
FF5	17	0.929

LS portfolios by cluster

$\# t(\alpha_{LS}) < 1.96$	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Accruals (1)	1	1	1	1	1	1	1	1
Debt Issuance (1)	1	1	1	0	0	0	1	0
Investment (2)	1	1	0	0	0	0	2	0
Profit growth (2)	1	1	1	1	1	1	1	1
Profitability (1)	1	1	1	1	0	0	1	1
Quality (6)	5	3	6	0	1	3	6	3
Seasonality (2)	2	2	2	2	2	2	1	2
Size (2)	1	1	0	0	1	1	1	1
Value (12)	12	10	6	6	8	8	6	8
Total (out of 29)	25	21	18	11	14	16	20	17

$ \overline{\alpha_{LS}} $	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Accruals	0.020	0.032	0.274	0.455	0.441	0.301	0.166	0.253
Debt Issuance	0.554	0.811	0.405	0.967	1.017	0.888	0.387	0.997
Investment	0.954	0.951	0.971	1.233	1.253	1.183	0.582	1.127
Profit Growth	0.886	0.992	0.820	0.909	0.879	0.806	0.797	0.978
Profitability	0.328	0.648	0.205	0.640	1.032	0.815	0.039	0.766
Quality	0.484	0.864	0.177	0.821	1.066	0.895	0.441	0.901
Seasonality	0.400	0.368	0.458	0.451	0.517	0.557	0.428	0.493
Size	0.668	0.725	1.278	1.037	0.607	0.584	0.923	0.555
Value	0.957	0.688	1.179	0.946	0.862	0.863	1.046	1.101

Table 7: Digesting anomalies: Mainboard versus non-Mainboard stocks (continued)**Panel B: Non-Mainboard stocks****All 17 LS Portfolios**

Models	# $t(\alpha_{LS}) < 1.96$ out of 19	$ \alpha_{LS} $
IDX4	16	0.709
IDX3 size+value	15	0.858
IDX3 size+prof	8	1.352
CAPM	9	1.463
FF3	7	1.416
CAR4	6	1.498
Q4	14	1.129
FF5	11	1.232

LS Portfolios by Cluster

# $t(\alpha_{LS}) < 1.96$	IDX4	IDX3 size+value	IDX3 Size+prof	CAPM	FF3	CAR4	Q4	FF5
Profit growth (2)	2	2	1	2	2	1	2	2
Profitability (2)	2	2	2	2	1	1	2	2
Size (1)	0	0	0	0	0	0	0	0
Value (12)	12	11	5	5	4	4	10	7
Total (out of 17)	16	15	8	9	7	6	14	11

$ \alpha_{LS} $	IDX4	IDX3 size+value	IDX3 size+prof	CAPM	FF3	CAR4	Q4	FF5
Profit growth	0.964	1.018	1.235	0.910	0.808	1.239	1.106	0.954
Profitability	0.237	0.529	0.022	0.646	1.135	1.223	0.054	0.754
Size	2.405	2.521	3.021	2.856	2.391	2.463	2.440	2.318
Value	0.604	0.748	1.454	1.576	1.484	1.507	1.202	1.268

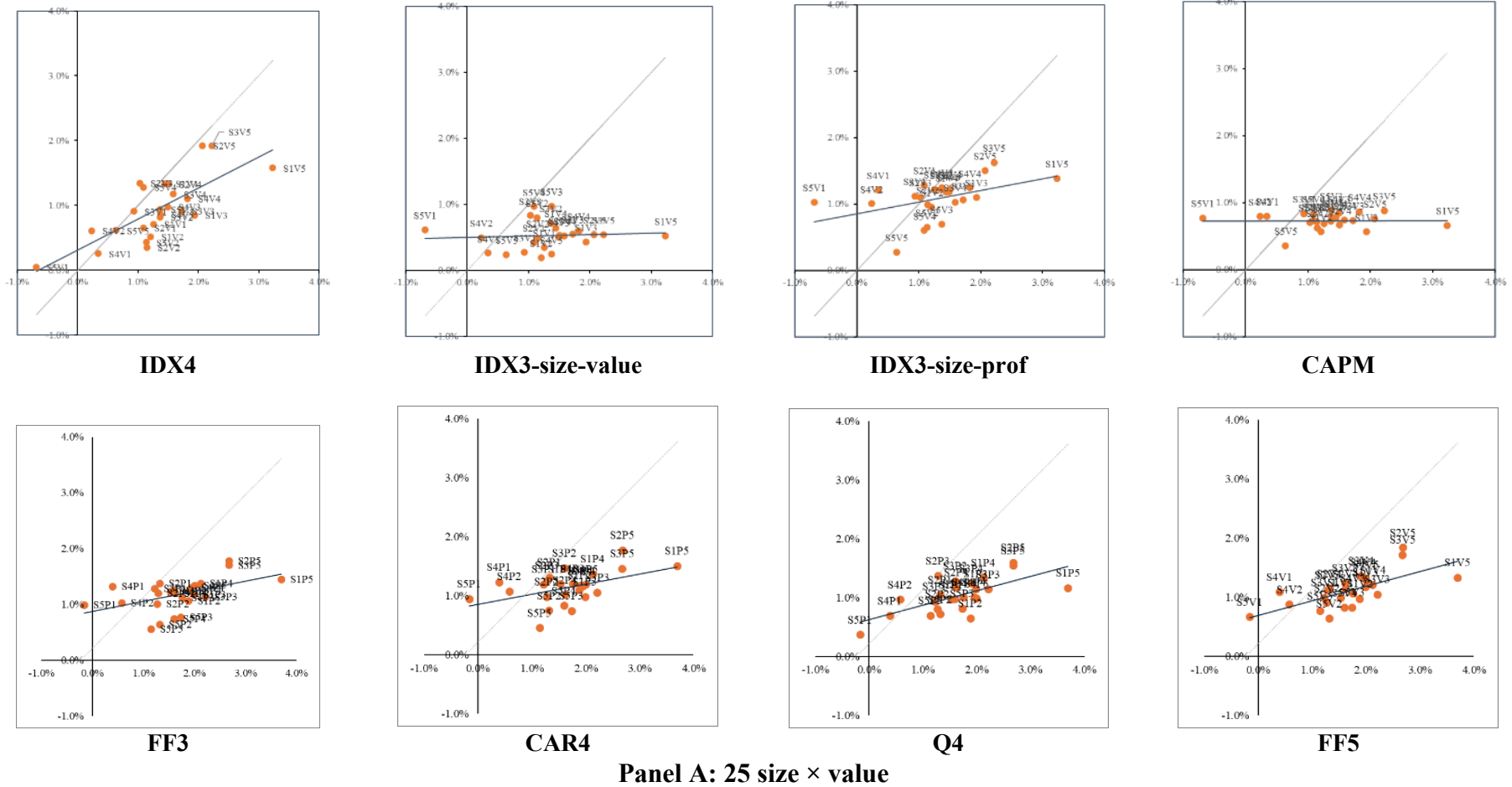
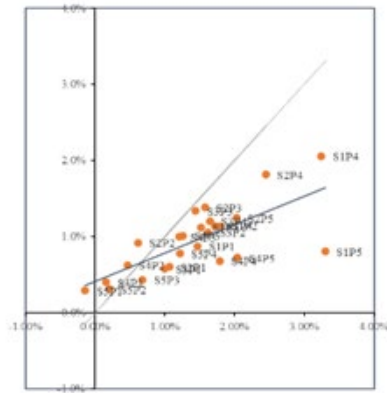
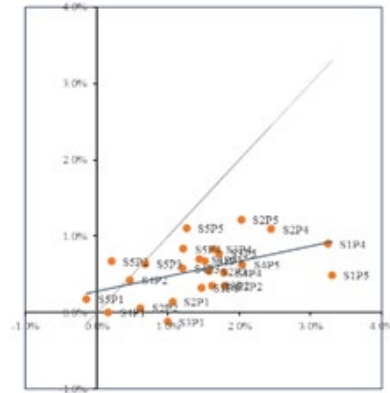


Figure 1: Model Implied Expected Return and Actual Return

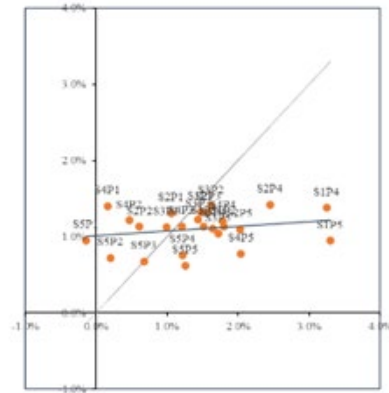
The figure plots scatter plots of model-implied portfolio returns against actual average returns on 25 portfolios sorted by size and profitability (Panel a), 25 portfolios sorted by size and value (Panel b), and 11 industry portfolios (Panel c). S1 and S5 represent the smallest and largest stocks. P1 and P5 represent the least and most profitable stocks. V1 and V5 represent stocks with the lowest and highest market valuation ratio (growth and value).



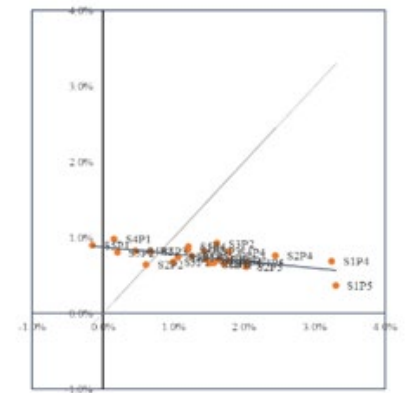
IDX4



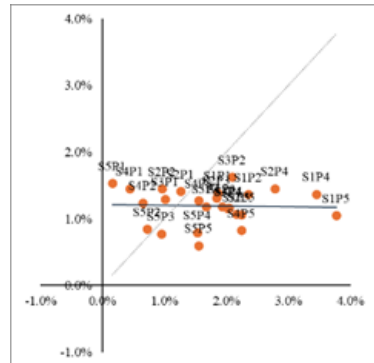
IDX3-size-value



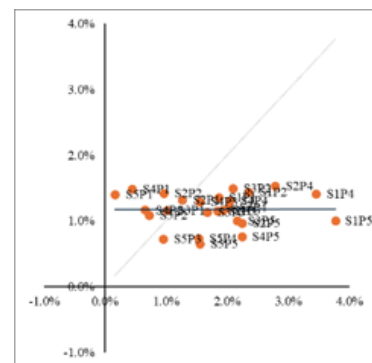
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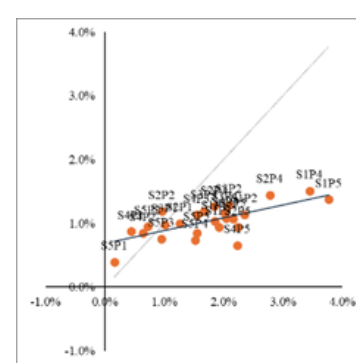
CAPM



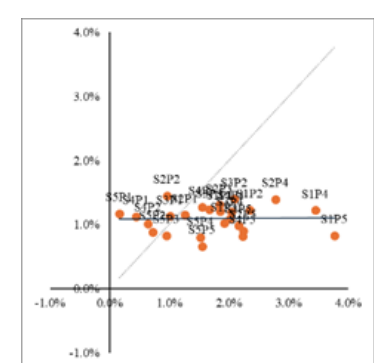
FF3



CAR4

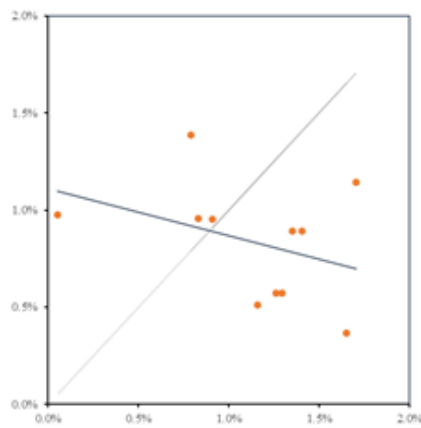


Q4

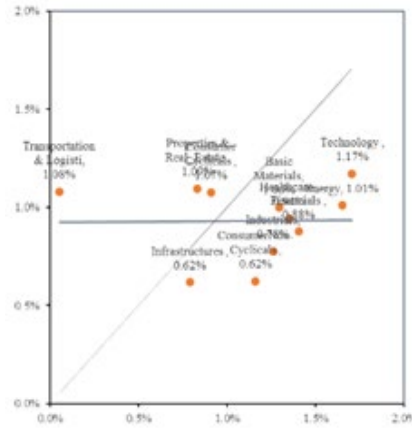


FF5

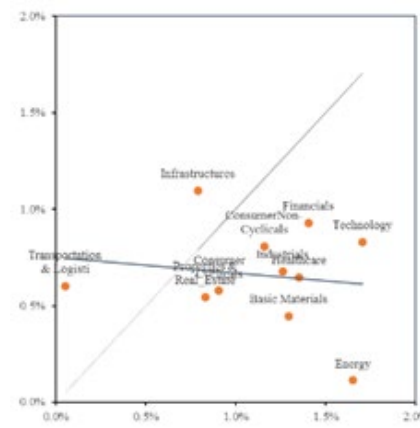
Panel B: 25 size $\times$ profitability



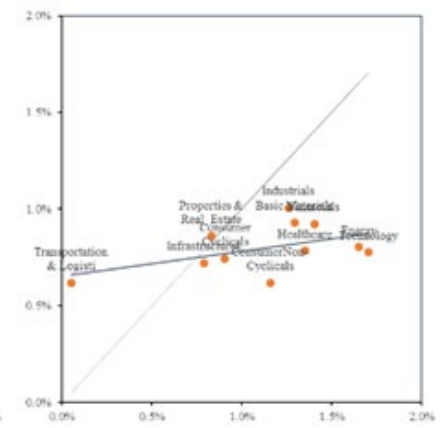
IDX4



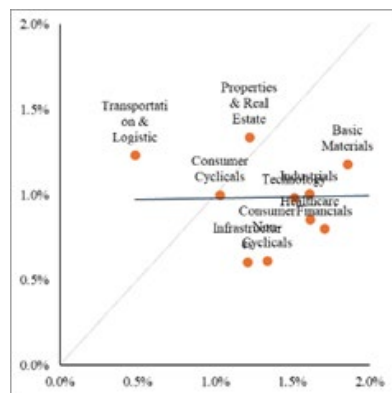
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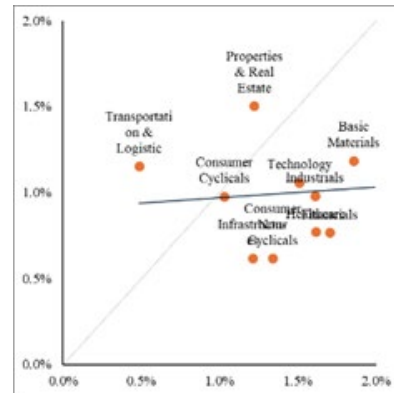
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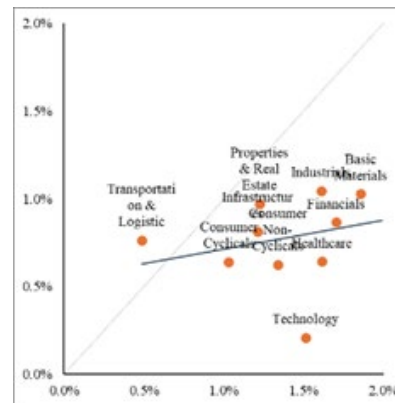
CAPM



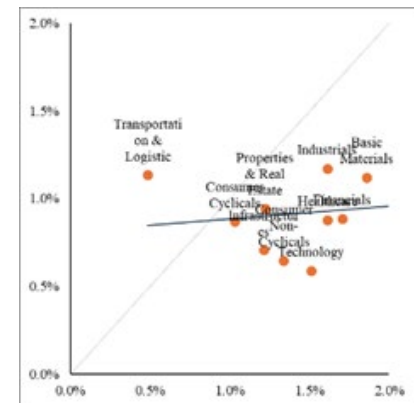
FF3



CAR4



Q4



FF5

Panel C: Industry portfolios

Internet Appendix

“A Four-Factor Model for the Indonesia Stock Market”

NOT INTENDED FOR PUBLICATION

Appendix A: Definitions of Firm Characteristics

Appendix B: Additional Tables

Table IA1: Correlation Matrix of Value and Profitability Risk Factors

Table IA2: Out-of-Sample R^2 to Explain IDX Equity Portfolios Sorted by Firm Characteristics

Table IA3: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Full Sample

Table IA4: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Subsamples

Appendix A: Definitions of Firm Characteristics

Cluster	Abbreviation	Description	Reference Paper
Accruals	cowc_gr1a	Change in current operating working capital	Richardson et al. (2005)
	oaccruals_at	Operating accruals	Sloan (1996)
	oaccruals_ni	Percent operating accruals	Hafzalla Lundholm and Van Winkle (2011)
	seas_16_20na	Years 16-20 lagged returns, nonannual	Heston and Sadka (2008)
	taccruals_at	Total accruals	Richardson et al. (2005)
Debt Issuance	taccruals_ni	Percent total accruals	Hafzalla Lundholm and Van Winkle (2011)
	capex_abn	Abnormal corporate investment	Titman Wei and Xie (2004)
	debt_gr3	Growth in book debt (3 years)	Lyandres Sun and Zhang (2008)
	fml_gr1a	Change in financial liabilities	Richardson et al. (2005)
	ncol_gr1a	Change in noncurrent operating liabilities	Richardson et al. (2005)
	nfna_gr1a	Change in net financial assets	Richardson et al. (2005)
	ni_ar1	Earnings persistence	Francis et al. (2004)
	noa_at	Net operating assets	Hirshleifer et al. (2004)
Investment	aliq_at	Liquidity of book assets	Ortiz-Molina and Phillips (2014)
	at_gr1	Asset Growth	Cooper Gulen and Schill (2008)
	be_gr1a	Change in common equity	Richardson et al. (2005)
	capx_gr1	CAPEX growth (1 year)	Xie (2001)
	capx_gr2	CAPEX growth (2 years)	Anderson and Garcia-Feijoo (2006)
	capx_gr3	CAPEX growth (3 years)	Anderson and Garcia-Feijoo (2006)
	coa_gr1a	Change in current operating assets	Richardson et al. (2005)
	col_gr1a	Change in current operating liabilities	Richardson et al. (2005)
	emp_gr1	Hiring rate	Belo Lin and Bazdresch (2014)
	inv_gr1	Inventory growth	Belo and Lin (2011)
	inv_gr1a	Inventory change	Thomas and Zhang (2002)
	lnoa_gr1a	Change in long-term net operating assets	Fairfield Whisenant and Yohn (2003)
	mispricing_mgmt	Mispricing factor: Management	Stambaugh and Yuan (2016)
	ncoa_gr1a	Change in noncurrent operating assets	Richardson et al. (2005)
	nncoa_gr1a	Change in net noncurrent operating assets	Richardson et al. (2005)
	noa_gr1a	Change in net operating assets	Hirshleifer et al. (2004)
	ppeinv_gr1a	Change PPE and Inventory	Lyandres Sun and Zhang (2008)
	ret_60_12	Long-term reversal	De Bondt and Thaler (1985)
	sale_gr1	Sales Growth (1 year)	Lakonishok Shleifer and Vishny (1994)
	sale_gr3	Sales Growth (3 years)	Lakonishok Shleifer and Vishny (1994)
	saleq_gr1	Sales growth (1 quarter)	Lakonishok Shleifer and Vishny (1994)
	seas_2_5na	Years 2-5 lagged returns, nonannual	Heston and Sadka (2008)

Appendix A (continued)

Cluster	Abbreviation	Brief Description	Reference Paper
Low Leverage	age	Firm age	Jiang Lee and Zhang (2005)
	aliq_mat	Liquidity of market assets	Ortiz-Molina and Phillips (2014)
	at_be	Book leverage	Fama and French (1992)
	bidaskhl_21d	The high-low bid-ask spread	Corwin and Schultz (2012)
	cash_at	Cash-to-assets	Palazzo (2012)
	netdebt_me	Net debt-to-price	Penman Richardson and Tuna (2007)
	ni_ivol	Earnings volatility	Francis et al. (2004)
	rd_sale	R&D-to-sales	Chan Lakonishok and Sougiannis (2001)
	rd5_at	R&D capital-to-book assets	Li (2011)
	tangibility	Asset tangibility	Hahn and Lee (2009)
	z_score	Altman Z-score	Dichev (1998)
Low Risk	beta_60m	Market Beta	Fama and MacBeth (1973)
	beta_dimson_21d	Dimson beta	Dimson (1979)
	betabab_1260d	Frazzini-Pedersen market beta	Frazzini and Pedersen (2014)
	betadown_252d	Downside beta	Ang Chen and Xing (2006)
	earnings_variability	Earnings variability	Francis et al. (2004)
	ivol_capm_21d	Idiosyncratic volatility from the CAPM (21 days)	
	ivol_capm_252d	Idiosyncratic volatility from the CAPM (252 days)	Ali Hwang and Trombley (2003)
	ivol_ff3_21d	Idiosyncratic volatility from the Fama-French 3-factor model	Ang et al. (2006)
	ivol_hxz4_21d	Idiosyncratic volatility from the q-factor model	
	ocfq_saleq_std	Cash flow volatility	Huang (2009)
	rmax1_21d	Maximum daily return	Bali Cakici and Whitelaw (2011)
	rmax5_21d	Highest 5 days of return	Bali, Brown, Murray and Tang (2017)
	rvol_21d	Return volatility	Ang et al. (2006)
	seas_6_10na	Years 6-10 lagged returns, nonannual	Heston and Sadka (2008)
	turnover_126d	Share turnover	Datar Naik and Radcliffe (1998)
	zero_trades_126d	Number of zero trades with turnover as tiebreaker (6 months)	Liu (2006)
	zero_trades_21d	Number of zero trades with turnover as tiebreaker (1 month)	Liu (2006)
	zero_trades_252d	Number of zero trades with turnover as tiebreaker (12 months)	Liu (2006)
Momentum	prc_highprc_252d	Current price to high price over last year	George and Hwang (2004)
	resff3_12_1	Residual momentum t-12 to t-1	Blitz Huij and Martens (2011)
	resff3_6_1	Residual momentum t-6 to t-1	Blitz Huij and Martens (2011)
	ret_12_1	Price momentum t-12 to t-1	Jegadeesh and Titman (1993)
	ret_3_1	Price momentum t-3 to t-1	Jegadeesh and Titman (1993)
	ret_6_1	Price momentum t-6 to t-1	Jegadeesh and Titman (1993)
	ret_9_1	Price momentum t-9 to t-1	Jegadeesh and Titman (1993)
	seas_1_1na	Year 1-lagged return, nonannual	Heston and Sadka (2008)

Appendix A (continued)

Cluster	Abbreviation	Brief Description	Reference Paper
Profit Growth	dsale_dinv	Change sales minus change Inventory	Abarbanell and Bushee (1998)
	dsale_drec	Change sales minus change	Abarbanell and Bushee (1998)
	dsale_dsga	Change sales minus change SG&A	Abarbanell and Bushee (1998)
	niq_at_chg1	Change in quarterly return on assets	
	niq_be_chg1	Change in quarterly return on equity	
	niq_su	Standardized earnings surprise	Foster Olsen and Shevlin (1984)
	ocf_at_chg1	Change in operating cash flow to assets	Bouchard, Krüger, Landier and Thesmar (2019)
	ret_12_7	Price momentum t-12 to t-7	Novy-Marx (2012)
	sale_emp_gr1	Labor force efficiency	Abarbanell and Bushee (1998)
	saleq_su	Standardized Revenue surprise	Jegadeesh and Livnat (2006)
Profitability	seas_1_lan	Year 1-lagged return, annual	Heston and Sadka (2008)
	tax_gr1a	Tax expense surprise	Thomas and Zhang (2011)
	dolvol_var_126d	Coefficient of variation for dollar trading volume	Chordia Subrahmanyam and Anshuman (2001)
	ebit_bev	Return on net operating assets	Soliman (2008)
	ebit_sale	Profit margin	Soliman (2008)
	f_score	Pitroski F-score	Pitroski (2000)
	ni_be	Return on equity	Haugen and Baker (1996)
	niq_be	Quarterly return on equity	Hou Xue and Zhang (2015)
	o_score	Ohlson O-score	Dichev (1998)
	ocf_at	Operating cash flow to assets	Bouchard, Krüger, Landier and Thesmar (2019)
Quality	ope_be	Operating profits-to-book equity	Fama and French (2015)
	ope_bell	Operating profits-to-lagged book	
	turnover_var_126d	Coefficient of variation for share turnover	Chordia Subrahmanyam and Anshuman (2001)
	at_turnover	Capital turnover	Haugen and Baker (1996)
	cop_at	Cash-based operating profits-to-book assets	
	cop_atl1	Cash-based operating profits-to-lagged book assets	Ball et al. (2016)
	dgp_dsale	Change gross margin minus change sales	Abarbanell and Bushee (1998)
	gp_at	Gross profits-to-assets	Novy-Marx (2013)
	gp_atl1	Gross profits-to-lagged assets	
	mispricing_perf	Mispricing factor: Performance	Stambaugh and Yuan (2016)
	ni_inc8q	Number of consecutive quarters with earnings increases	Barth Elliott and Finn (1999)
	niq_at	Quarterly return on assets	Balakrishnan Bartov and Faurel (2010)
	op_at	Operating profits-to-book assets	
	op_atl1	Operating profits-to-lagged book assets	Ball et al. (2016)
	opex_at	Operating leverage	Novy-Marx (2011)
	qmj	Quality minus Junk: Composite	Assness, Frazzini and Pedersen (2018)
	qmj_growth	Quality minus Junk: Growth	Assness, Frazzini and Pedersen (2018)
	qmj_prof	Quality minus Junk: Profitability	Assness, Frazzini and Pedersen (2018)
	qmj_safety	Quality minus Junk: Safety	Assness, Frazzini and Pedersen (2018)
	sale_bev	Asset turnover	Soliman (2008)

Appendix A (continued)

Cluster	Abbreviation	Brief Description	Reference Paper
Seasonality	corr_1260d	Market correlation	Assness, Frazzini, Gormsen, Pedersen (2020)
	coskew_21d	Coskewness	Harvey and Siddique (2000)
	dbnetis_at	Net debt issuance	Bradshaw Richardson and Sloan (2006)
	kz_index	Kaplan-Zingales index	Lamont Polk and Saa-Requejo (2001)
	lti_gr1a	Change in long-term investments	Richardson et al. (2005)
	pi_nix	Taxable income-to-book income	Lev and Nissim (2004)
	seas_11_15an	Years 11-15 lagged returns, annual	Heston and Sadka (2008)
	seas_11_15na	Years 11-15 lagged returns, nonannual	Heston and Sadka (2008)
	seas_16_20an	Years 16-20 lagged returns, annual	Heston and Sadka (2008)
	seas_2_5an	Years 2-5 lagged returns, annual	Heston and Sadka (2008)
	seas_6_10an	Years 6-10 lagged returns, annual	Heston and Sadka (2008)
Short-Term Reversal	sti_gr1a	Change in short-term investments	Richardson et al. (2005)
	iskew_capm_21d	Idiosyncratic skewness from the CAPM	
	iskew_ff3_21d	Idiosyncratic skewness from the Fama-French 3-factor model	Bali Engle and Murray (2016)
	iskew_hxz4_21d	Idiosyncratic skewness from the q-factor model	
	ret_1_0	Short-term reversal	Jegadeesh (1990)
	rmax5_rvol_21d	Highest 5 days of return scaled by volatility	Assness, Frazzini, Gormsen, Pedersen (2020)
Size	rskew_21d	Total skewness	Bali Engle and Murray (2016)
	ami_126d	Amihud Measure	Amihud (2002)
	dolvol_126d	Dollar trading volume	Brennan Chordia and Subrahmanyam (1998)
	market_equity	Market Equity	Banz (1981)
	prc	Price per share	Miller and Scholes (1982)
Value	rd_me	R&D-to-market	Chan Lakonishok and Sougiannis (2001)
	at_me	Assets-to-market	Fama and French (1992)
	be_me	Book-to-market equity	Rosenberg Reid and Lanstein (1985)
	bev_mev	Book-to-market enterprise value	Penman Richardson and Tuna (2007)
	chcsho_12m	Net stock issues	Pontiff and Woodgate (2008)
	debt_me	Debt-to-market	Bhandari (1988)
	div12m_me	Dividend yield	Litzenberger and Ramaswamy (1979)
	ebitda_mev	Ebitda-to-market enterprise value	Loughran and Wellman (2011)
	eq_dur	Equity duration	Dechow Sloan and Soliman (2004)
	eqnetis_at	Net equity issuance	Bradshaw Richardson and Sloan (2006)
	eqnp0_12m	Equity net payout	Daniel and Titman (2006)
	eqnp0_me	Net payout yield	Boudoukh et al. (2007)
	eqpo_me	Payout yield	Boudoukh et al. (2007)
	fcf_me	Free cash flow-to-price	Lakonishok Shleifer and Vishny (1994)
	ival_me	Intrinsic value-to-price	Frankel and Lee (1998)
	netis_at	Net total issuance	Bradshaw Richardson and Sloan (2006)
	ni_me	Earnings-to-price	Basu (1983)
	ocf_me	Operating cash flow-to-price	Desai Rajgopal and Venkatachalam (2004)
	sale_me	Sales-to-price	Barbee Mukherji and Raines (1996)

Table IA1: Correlation Matrix of Value and Profitability Risk Factors

The table reports pairwise correlations between value and profitability factors formed on different value and profitability variables.

		RMW							HML												
		<i>ebit_bev</i>	<i>ebit_sale</i>	<i>ni_be</i>	<i>niq_be</i>	<i>ocf_at</i>	<i>ope_be</i>	<i>ope_bell</i>	<i>at_me</i>	<i>be_me</i>	<i>div12m_me</i>	<i>ebitda_mev</i>	<i>eq_dur</i>	<i>eqnetis_at</i>	<i>eqnpo_me</i>	<i>fcf_me</i>	<i>ival_me</i>	<i>netis_at</i>	<i>ni_me</i>	<i>ocf_me</i>	<i>sale_me</i>
RMW	<i>ebit_bev</i>	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	<i>ebit_sale</i>	0.59	1.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0
	<i>ni_be</i>	0.73	0.51	1.00	0.00	0.00	0.00	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0
	<i>niq_be</i>	0.40	0.17	0.43	1.00	0.00	0.00	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0
	<i>ocf_at</i>	0.56	0.38	0.51	0.22	1.00	0.00	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0
	<i>ope_be</i>	0.65	0.50	0.63	0.46	0.42	1.00	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0
	<i>ope_bell</i>	0.54	0.49	0.53	0.41	0.50	0.76	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HML	<i>at_me</i>	-0.53	-0.39	-0.34	-0.14	-0.41	-0.39	-0.42	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>be_me</i>	-0.49	-0.41	-0.38	-0.27	-0.32	-0.43	-0.34	0.66	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>div12m_me</i>	-0.06	-0.04	0.09	0.14	-0.02	-0.02	-0.06	0.25	0.23	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>ebitda_mev</i>	-0.17	-0.05	-0.05	-0.09	-0.06	-0.05	-0.15	0.71	0.48	0.37	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>eq_dur</i>	-0.06	-0.07	-0.08	-0.14	-0.02	-0.18	-0.16	-0.07	0.21	-0.04	0.04	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>eqnetis_at</i>	0.01	0.03	0.01	-0.11	0.03	-0.01	0.04	-0.03	-0.02	0.04	0.03	0.58	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>eqnpo_me</i>	-0.08	-0.04	0.07	0.09	0.11	0.07	0.00	0.23	0.14	0.18	0.17	-0.07	-0.02	1.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>fcf_me</i>	0.07	0.05	-0.03	0.21	0.25	0.06	-0.02	0.02	-0.01	-0.22	-0.03	-0.17	-0.11	0.05	1.00	0.00	0.00	0.00	0.00	0.00
	<i>ival_me</i>	0.00	-0.06	0.05	-0.11	-0.01	-0.07	0.02	0.34	0.27	0.29	0.49	-0.12	-0.02	0.01	-0.04	1.00	0.00	0.00	0.00	0.00
	<i>netis_at</i>	-0.05	-0.09	-0.08	0.17	0.01	-0.18	-0.15	-0.06	-0.06	-0.04	-0.02	0.98	0.59	0.12	0.44	-0.15	1.00	0.00	0.00	0.00
	<i>ni_me</i>	0.48	0.43	0.50	0.02	0.31	0.43	0.32	-0.24	-0.19	0.23	0.23	-0.07	-0.07	-0.08	-0.13	0.24	-0.10	1.00	0.00	0.00
	<i>ocf_me</i>	0.03	0.11	0.16	0.06	0.45	0.13	0.12	0.39	0.36	0.25	0.59	-0.09	-0.02	0.26	0.21	0.33	-0.08	0.23	1.00	0.00
	<i>sale_me</i>	-0.34	-0.43	-0.16	-0.07	-0.17	-0.14	-0.17	0.73	0.55	0.22	0.65	0.04	0.01	0.20	-0.13	0.34	0.03	-0.07	0.44	1.00

Table IA2: Out-of-Sample R^2

The table reports out-of-sample predictive and total R^2 , which are proportion of total variations in returns on 32 sets of test portfolio sorted by anomaly variables explained by the product of factor loadings and average factor returns and product of risk exposures and realizations of risk factors. Risk exposures and average risk premiums are estimated strictly out-of-sample. R^2 are reported in percentage.

Char	Predictive R^2							
	CAPM	IDX3-size-value	IDX3-size-prof	IDX4	FF3	Car4	Q4	FF5
<i>oaccruals_ni</i>	1.91	1.39	1.92	2.45	-0.98	-1.28	-0.60	-1.06
<i>inv_gr1</i>	2.03	2.09	2.10	2.31	-1.01	-1.11	-0.84	-1.37
<i>rd_sale</i>	-1.17	-2.59	-0.69	-1.48	-1.28	-1.18	-3.41	-1.69
<i>dsale_dinv</i>	2.16	2.21	2.38	2.51	-0.70	-0.89	-0.86	-0.97
<i>tax_gr1a</i>	1.93	2.01	2.01	2.03	-0.92	-1.11	-0.76	-1.14
<i>ebit_bev</i>	0.73	0.77	0.29	1.24	-2.75	-2.74	-1.45	-2.39
<i>ebit_sale</i>	1.96	1.19	1.91	2.20	-1.21	-1.39	-0.65	-1.13
<i>ocf_at</i>	1.34	1.46	0.90	1.97	-1.66	-1.86	-0.91	-1.29
<i>cop_atl1</i>	1.18	1.33	1.21	1.51	-1.56	-1.73	-0.86	-1.62
<i>sale_bev</i>	1.45	1.62	1.47	2.04	-1.47	-1.37	-1.05	-1.36
<i>qmj_prof</i>	1.04	0.76	0.72	1.09	-1.70	-1.78	-1.29	-1.25
<i>qmj_growth</i>	1.54	2.14	1.70	2.37	-0.65	-0.37	-0.48	-0.73
<i>gp_at</i>	1.65	1.32	1.33	1.95	-1.96	-2.09	-1.11	-1.95
<i>gp_atl1</i>	1.71	1.22	1.28	2.20	-2.32	-2.43	-1.27	-2.18
<i>op_atl1</i>	2.11	1.85	2.16	2.44	-0.53	-0.68	-0.08	-0.53
<i>op_at</i>	2.03	1.82	2.00	2.15	-0.86	-0.94	-0.45	-0.88
<i>cop_at</i>	1.32	1.51	1.21	1.77	-1.72	-1.98	-1.05	-1.84
<i>seas_2_5an</i>	1.71	1.70	1.92	1.96	-1.69	-1.73	-1.37	-1.89
<i>dbnetis_at</i>	1.85	2.02	1.92	2.66	-1.11	-1.15	-0.81	-1.19
<i>sale_me</i>	2.77	1.72	3.69	3.47	1.08	0.92	0.49	1.08
<i>be_me</i>	2.02	0.81	2.45	2.44	0.09	-0.14	-0.65	0.17
<i>netis_at</i>	1.15	1.30	1.20	1.95	-0.48	-0.72	-0.65	-0.63
<i>ni_me</i>	2.22	1.77	2.46	2.48	-0.20	-0.29	0.05	-0.13
<i>ival_me</i>	1.97	1.54	2.39	2.40	-0.13	-0.25	-0.73	-0.31
<i>fcf_me</i>	2.01	1.61	2.15	2.84	-0.72	-0.65	-0.24	-0.89
<i>eqnpo_12m</i>	1.03	0.74	0.86	1.41	-2.01	-2.21	-1.40	-1.87
<i>eqnetis_at</i>	-0.87	-0.31	-2.32	-0.90	-1.65	-1.28	-0.82	-2.18
<i>eq_dur</i>	2.93	2.04	3.60	3.65	0.58	0.35	0.00	0.82
<i>ebitda_mev</i>	2.04	1.88	2.61	2.98	0.31	0.03	0.07	0.36
<i>ocf_me</i>	1.50	1.44	1.45	2.60	-1.17	-1.42	-0.70	-0.96
<i>at_me</i>	1.73	1.76	2.04	2.54	-0.03	-0.17	-0.21	-0.12
<i>me</i>	3.31	1.33	4.27	3.96	1.15	1.04	1.03	1.58

Table IA2: Out-of-Sample R^2

<i>Total R2</i>								
Char	CAPM	IDX3- size- value	IDX3- size-prof	IDX4	FF3	Car4	Q4	FF5
<i>oaccruals_ni</i>	64.44%	64.56%	65.63%	65.82%	66.48%	65.40%	66.03%	64.83%
<i>inv_gr1</i>	65.30%	64.94%	64.46%	64.57%	64.07%	63.92%	63.76%	64.77%
<i>rd_sale</i>	22.20%	21.37%	22.01%	21.13%	19.88%	19.18%	16.57%	15.19%
<i>dsale_dinv</i>	62.69%	62.77%	62.80%	63.04%	63.28%	62.57%	63.17%	63.32%
<i>tax_gr1a</i>	72.38%	72.11%	72.59%	72.96%	71.70%	71.99%	71.02%	71.30%
<i>ebit_bev</i>	71.24%	79.66%	74.85%	80.21%	75.37%	74.85%	75.23%	75.53%
<i>ebit_sale</i>	68.36%	75.63%	73.35%	76.82%	73.85%	72.87%	71.32%	71.45%
<i>ocf_at</i>	71.78%	75.49%	73.88%	76.81%	74.28%	73.04%	73.62%	73.94%
<i>cop_atl1</i>	62.12%	66.91%	63.42%	67.54%	66.81%	66.08%	65.91%	64.36%
<i>sale_bev</i>	74.15%	78.32%	76.08%	78.73%	75.82%	75.57%	75.81%	75.01%
<i>qmj_prof</i>	69.50%	73.41%	72.59%	76.60%	72.92%	72.15%	72.64%	74.74%
<i>qmj_growth</i>	70.48%	71.63%	71.04%	70.66%	71.40%	70.96%	71.84%	71.30%
<i>gp_at</i>	63.41%	69.49%	65.13%	68.56%	66.75%	66.88%	67.24%	66.88%
<i>gp_atl1</i>	63.36%	68.69%	65.53%	68.66%	68.02%	68.09%	68.30%	68.11%
<i>op_atl1</i>	71.34%	72.12%	72.33%	73.69%	72.66%	72.22%	72.26%	73.27%
<i>op_at</i>	73.22%	74.39%	74.15%	75.54%	74.26%	73.36%	73.79%	75.44%
<i>cop_at</i>	64.19%	69.48%	65.74%	70.18%	69.23%	69.22%	69.15%	68.29%
<i>seas_2_5an</i>	70.13%	70.82%	70.77%	69.50%	70.66%	69.31%	68.59%	69.40%
<i>dbnetis_at</i>	72.47%	73.64%	73.19%	74.15%	72.47%	71.29%	73.20%	72.50%
<i>sale_me</i>	71.28%	73.21%	77.26%	78.87%	80.56%	81.19%	76.17%	77.38%
<i>be_me</i>	70.82%	72.00%	79.21%	80.47%	84.85%	84.76%	77.32%	82.90%
<i>netis_at</i>	45.17%	45.40%	44.29%	44.89%	43.35%	42.41%	43.67%	42.62%
<i>ni_me</i>	69.93%	72.64%	72.16%	75.38%	71.27%	70.17%	70.22%	71.93%
<i>ival_me</i>	62.58%	62.68%	63.21%	65.39%	63.77%	62.99%	58.96%	63.04%
<i>fcf_me</i>	61.96%	65.81%	63.26%	67.16%	66.17%	66.82%	66.39%	64.85%
<i>eqnpo_12m</i>	72.78%	76.14%	75.15%	76.70%	74.61%	73.82%	74.25%	73.05%
<i>eqnetis_at</i>	27.24%	29.80%	26.51%	26.19%	37.57%	37.85%	38.49%	35.93%
<i>eq_dur</i>	72.93%	74.28%	7.89E-01	79.96%	80.90%	80.65%	75.89%	78.22%
<i>ebitda_mev</i>	72.03%	72.42%	74.40%	76.70%	75.73%	75.55%	72.45%	74.99%
<i>ocf_me</i>	68.25%	71.87%	70.47%	77.07%	70.93%	70.44%	72.22%	72.42%
<i>at_me</i>	73.86%	73.95%	75.88%	75.93%	76.46%	75.50%	73.15%	75.58%
<i>me</i>	67.64%	77.51%	83.97%	86.68%	84.17%	83.57%	80.49%	77.99%

Table IA3: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Full sample

The table reports alphas of long-short portfolios buying the most undervalued quintile portfolio and selling the most overvalued quintile portfolio sorted by different firm characteristics relative to four candidate models. Portfolio alphas are reported in percentage. t-statistics to test the null hypothesis of zero portfolio alphas are reported in parenthesis.

Cluster	Char	CAPM		IDX3-size-value		IDX3-size-prof		IDX4		FF3		Car4		Q4		FF5	
		α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$
Accruals	<i>oaccruals_ni</i>	1.019	(2.24)	0.457	(0.97)	0.693	(1.26)	0.628	(1.21)	1.103	(2.56)	1.013	(2.30)	0.698	(1.30)	0.939	(1.98)
Investment	<i>inv_gr1</i>	0.210	(0.59)	0.040	(0.09)	0.642	(1.86)	0.121	(0.23)	0.178	(0.45)	0.364	(0.91)	0.247	(0.57)	0.082	(0.20)
Low Leverage	<i>rd_sale</i>	1.198	(1.72)	1.874	(2.79)	2.124	(2.07)	1.873	(2.29)	1.225	(1.87)	1.604	(2.37)	1.811	(1.84)	1.611	(2.31)
Profit Growth	<i>dsale_dinv</i>	1.094	(2.63)	1.142	(2.17)	1.436	(3.28)	1.017	(1.69)	1.045	(2.43)	1.214	(2.78)	1.214	(2.20)	1.165	(2.45)
Profitability	<i>tax_gr1a</i>	0.639	(1.67)	0.908	(2.04)	0.498	(1.49)	0.595	(1.60)	0.744	(2.13)	0.682	(1.82)	0.460	(1.10)	0.839	(2.10)
	<i>ebit_bev</i>	1.393	(2.37)	1.191	(2.38)	0.032	(0.09)	0.504	(1.28)	1.860	(4.02)	1.596	(3.81)	0.773	(1.92)	1.279	(2.71)
	<i>ebit_sale</i>	0.849	(1.59)	0.650	(1.70)	-0.090	(0.23)	0.410	(1.03)	1.265	(3.55)	1.226	(3.48)	0.899	(2.59)	0.887	(2.26)
Quality	<i>ocf_at</i>	0.832	(1.45)	0.507	(0.96)	-0.128	(0.29)	0.174	(0.37)	1.239	(2.70)	1.105	(2.58)	0.144	(0.40)	0.705	(1.59)
	<i>cop_at1</i>	1.198	(2.42)	1.005	(2.21)	0.166	(0.42)	0.337	(0.86)	1.534	(3.59)	1.382	(3.36)	0.686	(1.76)	1.163	(2.58)
	<i>sale_bev</i>	0.947	(2.32)	0.912	(1.98)	0.032	(0.11)	0.307	(0.81)	1.311	(3.74)	1.229	(3.66)	0.333	(0.93)	1.003	(2.51)
	<i>qmj_prof</i>	0.636	(1.19)	0.462	(0.87)	-0.425	(0.89)	-0.180	(0.38)	1.111	(2.56)	0.939	(2.24)	-0.277	(0.60)	0.341	(0.79)
	<i>qmj_growth</i>	0.343	(0.88)	0.458	(1.08)	-0.214	(0.60)	0.246	(0.61)	0.499	(1.32)	0.342	(0.94)	0.218	(0.53)	0.402	(1.11)
	<i>gp_at</i>	1.363	(2.69)	1.482	(2.98)	0.013	(0.04)	0.525	(1.44)	1.773	(3.72)	1.537	(3.39)	0.600	(1.47)	1.453	(3.12)
	<i>gp_at1</i>	1.255	(2.71)	1.483	(3.10)	0.079	(0.22)	0.516	(1.38)	1.615	(3.66)	1.398	(3.34)	0.612	(1.54)	1.323	(3.09)
Seasonality	<i>op_at1</i>	-0.115	(0.26)	-0.278	(0.63)	-0.819	(1.99)	-0.804	(1.87)	0.071	(0.17)	0.045	(0.11)	-0.684	(1.51)	-0.516	(1.25)
	<i>op_at</i>	0.047	(0.11)	-0.106	(0.24)	-0.447	(1.26)	-0.576	(1.47)	0.194	(0.49)	0.288	(0.74)	-0.412	(1.13)	-0.346	(0.90)
	<i>cop_at</i>	1.079	(2.16)	0.775	(1.67)	-0.031	(0.09)	0.105	(0.28)	1.420	(3.30)	1.273	(3.12)	0.592	(1.55)	0.959	(2.22)
	<i>seas_2_San</i>	0.112	(0.23)	0.565	(1.01)	0.149	(0.29)	-0.039	(0.08)	0.208	(0.41)	0.244	(0.45)	0.095	(0.19)	0.094	(0.19)
	<i>dbnetis_at</i>	0.162	(0.43)	-0.336	(0.83)	-0.184	(0.51)	-0.643	(1.74)	0.229	(0.57)	0.194	(0.52)	-0.306	(0.88)	-0.121	(0.32)
Value	<i>sale_me</i>	0.937	(2.31)	-0.074	(0.25)	1.225	(3.20)	0.140	(0.54)	0.400	(1.42)	0.503	(1.70)	0.912	(2.46)	0.246	(0.83)
	<i>be_me</i>	0.393	(1.01)	-0.513	(1.39)	1.351	(3.25)	0.063	(0.18)	-0.359	(1.63)	-0.134	(0.60)	1.015	(2.69)	-0.187	(0.74)
	<i>netis_at</i>	0.683	(0.88)	0.145	(0.19)	0.173	(0.19)	-0.048	(0.05)	0.787	(1.00)	0.400	(0.51)	-0.380	(0.51)	0.213	(0.27)
	<i>ni_me</i>	1.429	(3.21)	0.720	(2.33)	1.352	(3.50)	0.687	(1.93)	1.288	(3.26)	1.290	(3.33)	1.381	(3.03)	0.935	(2.39)
	<i>ival_me</i>	0.984	(1.82)	-0.063	(0.12)	1.706	(3.13)	0.504	(0.98)	0.459	(0.94)	0.794	(1.72)	1.543	(2.92)	0.154	(0.32)
	<i>fcf_me</i>	1.417	(2.98)	1.097	(2.15)	0.493	(1.03)	0.704	(1.33)	1.719	(3.91)	1.415	(3.37)	0.570	(1.04)	1.347	(2.80)
	<i>eqnpo_12m</i>	0.939	(2.18)	0.521	(1.30)	0.465	(1.23)	0.331	(1.01)	1.148	(3.01)	1.233	(3.07)	0.459	(1.36)	0.684	(1.66)
	<i>eqnetis_at</i>	2.647	(3.71)	2.507	(3.27)	1.642	(1.88)	1.834	(1.98)	2.813	(4.01)	2.146	(2.94)	2.099	(2.34)	2.402	(3.18)
	<i>eq_dur</i>	0.844	(2.03)	-0.143	(0.41)	1.402	(3.56)	0.373	(0.99)	0.277	(0.80)	0.401	(1.19)	1.182	(2.89)	0.279	(0.80)
	<i>ebitda_mev</i>	1.092	(2.15)	-0.184	(0.46)	1.618	(3.33)	0.554	(1.15)	0.687	(1.68)	1.017	(2.59)	1.677	(3.05)	0.686	(1.49)
	<i>ocf_me</i>	1.574	(3.10)	0.187	(0.45)	1.326	(3.28)	0.432	(1.02)	1.497	(3.29)	1.658	(3.59)	1.313	(3.03)	0.897	(2.32)
	<i>at_me</i>	0.992	(2.31)	0.102	(0.29)	1.375	(3.12)	0.583	(1.35)	0.672	(2.04)	0.882	(2.46)	1.670	(3.35)	0.805	(2.25)
	<i>me</i>	0.925	(2.76)	0.479	(2.00)	1.196	(4.43)	0.599	(2.89)	0.509	(2.62)	0.565	(2.92)	0.651	(2.84)	0.500	(2.67)

Table IA4: Asset Pricing Test: Time-series GRS Tests

Cluster	Measure		IDX4	IDX3-value	IDX3-prof	CAPM	FF3	Car4	Q4	FF5
Accrual	<i>oaccruals</i>	GRS Stat	1.795	1.081	6.753	4.543	3.251	2.577	6.394	4.513
		<i>p</i> -value	0.114	0.371	0.000	0.001	0.007	0.027	0.000	0.001
		$ \alpha $	0.374	0.262	0.684	0.574	0.463	0.421	0.508	0.440
Investment	<i>inv_gr1</i>	GRS Stat	3.448	3.239	6.376	4.101	3.279	3.914	5.069	3.344
		<i>p</i> -value	0.005	0.007	0.000	0.001	0.007	0.002	0.000	0.006
		$ \alpha $	0.441	0.452	0.614	0.466	0.412	0.498	0.495	0.380
	<i>rd_sale</i>	GRS Stat	6.161	5.925	9.151	3.551	3.664	6.037	3.014	3.974
		<i>p</i> -value	0.000	0.000	0.000	0.004	0.003	0.000	0.012	0.002
		$ \alpha $	0.708	0.609	0.656	0.497	0.488	0.668	0.594	0.522
Profit growth	<i>dsale_din</i>	GRS Stat	6.421	4.593	7.151	6.659	6.123	6.622	6.841	4.726
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.555	0.441	0.596	0.645	0.606	0.646	0.601	0.461
	<i>tax_gr1a</i>	GRS Stat	7.324	8.706	9.458	10.323	10.033	9.876	19.012	18.984
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.383	0.387	0.509	0.4893	0.392	0.415	0.603	0.530
Profitability	<i>ebit_bev</i>	GRS Stat	14.020	14.257	15.747	13.66	20.714	20.950	23.156	21.994
		<i>p</i> -value	0.000	0.000	0.000	0	0.000	0.000	0.000	0.000
		$ \alpha $	0.440	0.540	0.560	0.4081	0.683	0.691	0.427	0.501
	<i>ebit_sale</i>	GRS Stat	12.996	14.159	20.224	18.052	19.587	19.630	20.742	22.406
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.442	0.329	0.670	0.5056	0.570	0.591	0.615	0.448
	<i>ocf_at</i>	GRS Stat	6.756	7.504	10.081	11.581	10.639	8.619	18.619	19.777
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.407	0.365	0.618	0.684	0.590	0.527	0.554	0.652
Quality	<i>cop_at11</i>	GRS Stat	2.463	3.252	6.401	6.655	7.054	7.254	6.582	8.921
		<i>p</i> -value	0.033	0.007	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.359	0.383	0.609	0.6475	0.593	0.580	0.480	0.488
	<i>cop_at</i>	GRS Stat	3.188	4.176	7.591	9.308	8.946	8.106	6.799	9.404
		<i>p</i> -value	0.008	0.001	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.315	0.351	0.640	0.6693	0.566	0.559	0.599	0.559
	<i>sale_bev</i>	GRS Stat	10.929	12.291	16.132	16.860	16.273	17.217	20.588	16.895
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.466	0.450	0.584	0.542	0.582	0.636	0.551	0.491
	<i>qmj_prof</i>	GRS Stat	7.540	9.047	11.768	13.022	13.519	13.563	17.287	21.285
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.372	0.447	0.478	0.5027	0.553	0.579	0.580	0.449
	<i>qmj_gro</i>	GRS Stat	4.850	5.113	8.022	8.598	7.836	7.556	14.916	11.802
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.646	0.626	0.851	0.7605	0.751	0.786	0.558	0.626
	<i>gp_at</i>	GRS Stat	5.600	7.617	7.174	9.158	8.926	9.552	7.697	9.572
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.577	0.654	0.595	0.706	0.623	0.681	0.545	0.621
	<i>gp_at11</i>	GRS Stat	2.750	2.072	5.222	4.386	3.831	4.428	7.030	7.701
		<i>p</i> -value	0.019	0.069	0.000	0.001	0.002	0.001	0.000	0.000
		$ \alpha $	0.509	0.423	0.691	0.6389	0.613	0.694	0.677	0.556
	<i>op_at11</i>	GRS Stat	9.781	8.635	13.531	10.917	10.612	10.357	18.136	18.456
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.520	0.477	0.638	0.553	0.521	0.509	0.660	0.530
	<i>op_at</i>	GRS Stat	11.444	11.751	14.070	14.810	15.561	14.195	19.372	19.143
		<i>p</i> -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		$ \alpha $	0.367	0.245	0.464	0.551	0.589	0.553	0.480	0.511

Table IA4: Continued

Cluster		CAPM	IDX3-	IDX3-	IDX4	FF3	Car4	O4	FF5
Seasonality	GRS Stat	4.947	3.325	7.598	3.695	3.488	3.319	25.227	8.974
	p-value	0.000	0.006	0.000	0.003	0.004	0.006	0.000	0.000
	$ \alpha $	0.496	0.488	0.559	0.470	0.426	0.441	0.699	0.440
	GRS Stat	16.317	12.412	18.389	15.207	15.395	14.183	16.170	18.152
	p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	0.622	0.579	0.600	0.516	0.653	0.653	0.538	0.472
Value	GRS Stat	11.360	11.135	17.289	7.434	9.710	11.344	26.254	25.595
	p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	0.983	0.330	1.346	0.408	0.458	0.576	0.886	0.639
	GRS Stat	14.539	11.157	20.336	9.438	13.357	14.653	21.771	23.536
	p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	1.025	0.351	1.560	0.473	0.376	0.482	0.814	0.639
	GRS Stat	4.525	4.629	4.125	3.630	4.842	3.367	2.051	3.475
	p-value	0.001	0.000	0.001	0.003	0.000	0.006	0.073	0.005
	$ \alpha $	0.605	0.695	0.639	0.586	0.642	0.538	0.478	0.544
	GRS Stat	3.551	1.969	9.274	1.307	2.008	2.347	18.480	17.281
	p-value	0.004	0.083	0.000	0.261	0.077	0.041	0.000	0.000
	$ \alpha $	0.765	0.301	1.024	0.287	0.565	0.520	0.737	0.608
	GRS Stat	4.390	1.531	4.935	1.206	3.004	3.664	7.391	5.900
	p-value	0.001	0.180	0.000	0.306	0.012	0.003	0.000	0.000
	$ \alpha $	0.673	0.237	0.737	0.234	0.460	0.538	0.666	0.506
	GRS Stat	7.980	3.339	10.783	3.464	7.076	6.522	6.788	7.484
	p-value	0.000	0.006	0.000	0.005	0.000	0.000	0.000	0.000
	$ \alpha $	0.888	0.586	1.069	0.548	0.821	0.744	0.659	0.671
	GRS Stat	16.768	11.828	22.450	14.624	15.305	15.802	57.011	34.715
	p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	0.437	0.343	0.499	0.416	0.451	0.523	0.552	0.413
	GRS Stat	5.073	5.532	9.399	1.649	1.889	1.327	0.883	2.847
	p-value	0.000	0.000	0.000	0.147	0.096	0.253	0.493	0.016
	$ \alpha $	0.491	0.563	0.712	0.369	0.366	0.317	0.323	0.506
	GRS Stat	7.461	2.101	13.764	3.867	6.115	7.705	22.094	17.253
	p-value	0.000	0.065	0.000	0.002	0.000	0.000	0.000	0.000
	$ \alpha $	1.011	0.364	1.299	0.480	0.618	0.614	0.898	0.700
	GRS Stat	5.380	4.182	9.464	2.598	3.890	4.714	21.658	18.435
	p-value	0.000	0.001	0.000	0.026	0.002	0.000	0.000	0.000
	$ \alpha $	0.999	0.253	1.170	0.257	0.517	0.637	0.728	0.618
	GRS Stat	7.497	4.283	8.368	2.932	5.981	4.999	22.835	23.120
	p-value	0.000	0.001	0.000	0.013	0.000	0.000	0.000	0.000
	$ \alpha $	1.002	0.279	0.905	0.225	0.737	0.700	0.647	0.751
	GRS Stat	10.867	9.908	15.952	7.109	9.220	10.754	18.709	20.488
	p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	0.737	0.427	1.090	0.322	0.451	0.559	0.753	0.594
Size	GRS Stat	77.457	91.814	111.000	74.878	79.516	82.961	357.000	136.000
	p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	$ \alpha $	1.031	0.525	1.360	0.611	0.422	0.512	0.836	0.585

Table IA5: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Subsamples

The table reports alphas of long-short portfolios buying the most undervalued quintile portfolio and selling the most overvalued quantile portfolio sorted by different firm characteristics relative to four candidate models within each subsample of stock. Panel A, B, C, and D reports results for Sharia stocks, non-Sharia stocks, stocks listed on the Main board, and stocks not listed on the Main board. Portfolio alphas are reported in percentage. *t*-statistics to test the null hypothesis of zero portfolio alphas are reported in parathesis.

Panel A: Sharia Stocks

Cluster	Char	CAPM		IDX3-size-value		IDX3-size-prof		IDX4		FF3		Car4		Q4		FF5	
		α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$
Low Risk	<i>betabab_1260d</i>	1.001	(1.80)	0.999	(1.98)	0.354	(0.79)	0.422	(1.03)	0.820	(1.62)	0.503	(1.06)	0.525	(1.01)	0.853	(1.76)
Profit Growth	<i>dsale_dinv</i>	1.196	(2.23)	1.481	(2.67)	1.177	(2.09)	1.385	(2.46)	1.246	(2.30)	1.306	(2.45)	1.175	(2.05)	1.438	(2.71)
	<i>niq_at_chg1</i>	0.557	(1.54)	0.788	(1.99)	0.328	(0.92)	0.485	(1.23)	0.539	(1.49)	0.334	(0.95)	0.214	(0.61)	0.581	(1.55)
Quality	<i>cop_at</i>	0.787	(1.41)	0.759	(1.32)	-0.170	(0.34)	0.214	(0.41)	1.115	(2.22)	0.907	(1.91)	0.218	(0.43)	0.829	(1.69)
Seasonality	<i>dbnetis_at</i>	0.384	(1.20)	0.097	(0.28)	0.275	(0.76)	0.229	(0.63)	0.455	(1.41)	0.423	(1.27)	0.124	(0.35)	0.377	(1.13)
Size	<i>me</i>	1.175	(3.24)	0.870	(2.43)	1.368	(3.99)	0.736	(2.60)	0.652	(2.51)	0.677	(2.50)	1.174	(3.36)	0.675	(2.37)
Value	<i>be_me</i>	0.621	(1.38)	-0.051	(0.10)	1.396	(3.27)	-0.107	(0.33)	-0.170	(0.62)	-0.112	(0.37)	0.604	(1.64)	-0.320	(1.05)
	<i>ebitda_mev</i>	1.130	(2.45)	0.020	(0.05)	1.463	(3.13)	0.446	(1.10)	0.868	(1.97)	1.126	(2.50)	1.277	(2.63)	0.715	(1.62)
	<i>eq_dur</i>	1.108	(2.49)	0.273	(0.67)	1.501	(3.52)	0.460	(1.26)	0.620	(1.62)	0.715	(1.89)	1.091	(2.75)	0.586	(1.52)
	<i>eqpo_me</i>	0.115	(0.10)	-2.568	(1.40)	1.403	(1.00)	-5.430	(1.00)	-1.216	(1.10)	-1.074	(0.78)	3.376	(2.82)	5.340	(2.71)
	<i>fcf_me</i>	1.519	(2.75)	1.191	(2.01)	0.867	(1.40)	1.112	(1.80)	1.613	(3.01)	1.382	(2.39)	0.656	(0.99)	1.480	(2.53)
	<i>ival_me</i>	0.369	(0.73)	-0.714	(1.38)	0.860	(1.67)	-0.271	(0.59)	0.027	(0.06)	0.345	(0.79)	0.750	(1.47)	-0.083	(0.19)
	<i>netis_at</i>	0.780	(1.42)	0.607	(1.10)	1.064	(1.69)	0.986	(1.69)	0.871	(1.59)	0.731	(1.29)	-0.006	(0.01)	0.650	(1.19)
	<i>ocf_me</i>	1.566	(3.10)	0.442	(0.89)	1.225	(2.38)	0.750	(1.44)	1.594	(3.15)	1.576	(3.14)	0.894	(1.98)	1.169	(2.32)
	<i>sale_me</i>	1.176	(2.36)	0.271	(0.64)	1.631	(3.23)	0.453	(1.19)	0.600	(1.53)	0.600	(1.44)	1.076	(2.42)	0.316	(0.81)

Table IA5: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Subsamples

Panel B: non-Sharia Stocks

Cluster	Char	CAPM		IDX3-size-value		IDX3-size-prof		IDX4		FF3		Car4		Q4		FF5	
		α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$
Accruals	<i>oaccruals_ni</i>	0.775	(1.41)	-0.082	(0.13)	0.822	(1.40)	0.225	(0.37)	0.939	(1.53)	1.310	(2.16)	0.890	(1.22)	0.747	(1.16)
Debt Issuance	<i>fnl_gr1a</i>	0.339	(0.51)	-0.072	(0.09)	0.255	(0.34)	-0.638	(0.91)	0.078	(0.11)	-0.211	(0.28)	0.016	(0.02)	-0.202	(0.29)
	<i>nfna_gr1a</i>	0.141	(0.19)	-0.411	(0.48)	0.089	(0.12)	-0.759	(1.01)	-0.129	(0.17)	-0.299	(0.37)	-0.111	(0.15)	-0.455	(0.61)
Investment	<i>mispricing_mgmt</i>	0.530	(0.70)	-0.317	(0.42)	-0.267	(0.34)	-0.669	(0.94)	0.663	(0.88)	0.190	(0.25)	-0.907	(1.17)	0.073	(0.10)
	<i>nncoa_gr1a</i>	0.175	(0.17)	-0.106	(0.08)	0.436	(0.39)	-0.272	(0.27)	-0.075	(0.07)	-0.347	(0.33)	-0.067	(0.06)	-0.619	(0.61)
	<i>noa_gr1a</i>	0.661	(0.74)	0.188	(0.19)	0.633	(0.79)	0.152	(0.20)	0.752	(0.88)	0.407	(0.45)	0.568	(0.53)	0.242	(0.27)
Momentum	<i>resff3_12_1</i>	0.615	(0.96)	0.140	(0.21)	0.192	(0.26)	0.025	(0.03)	0.586	(0.95)	0.101	(0.15)	-0.293	(0.37)	0.307	(0.49)
	<i>ret_3_1</i>	2.520	(3.31)	2.755	(2.95)	2.262	(2.79)	2.740	(2.74)	2.699	(3.48)	2.409	(2.97)	2.662	(2.85)	2.933	(3.47)
Profit Growth	<i>dsale_dinv</i>	0.635	(0.84)	0.742	(0.86)	1.296	(2.02)	0.642	(1.04)	0.466	(0.62)	0.814	(1.05)	1.017	(1.45)	0.513	(0.71)
	<i>ret_12_7</i>	0.263	(0.31)	0.681	(0.78)	-0.269	(0.31)	0.814	(0.83)	0.362	(0.46)	-0.222	(0.28)	-0.070	(0.07)	0.516	(0.63)
Profitability	<i>doivol_var_126d</i>	-0.904	(1.69)	-1.033	(1.36)	-1.779	(2.10)	-0.850	(1.41)	-0.349	(0.73)	-0.784	(1.31)	-1.063	(1.58)	-0.227	(0.44)
	<i>ni_be</i>	0.408	(0.57)	-0.768	(1.04)	-0.859	(1.02)	-0.835	(1.08)	0.813	(1.24)	0.344	(0.45)	-1.231	(1.53)	0.046	(0.07)
	<i>ocf_at</i>	1.263	(1.73)	0.099	(0.16)	0.464	(0.72)	0.291	(0.48)	1.789	(2.79)	1.723	(2.60)	0.451	(0.72)	1.383	(1.98)
	<i>ope_bell</i>	0.656	(1.01)	-0.437	(0.67)	0.040	(0.07)	-0.206	(0.35)	1.150	(1.97)	1.257	(2.08)	0.004	(0.01)	0.742	(1.18)
	<i>turnover_var_126d</i>	-0.989	(1.77)	-1.208	(1.63)	-1.850	(2.25)	-0.931	(1.57)	-0.528	(1.05)	-0.953	(1.57)	-1.215	(1.81)	-0.553	(0.98)
Quality	<i>cop_at</i>	1.341	(1.75)	0.222	(0.28)	0.232	(0.29)	0.505	(0.78)	1.905	(2.72)	1.809	(2.45)	0.537	(0.68)	1.661	(2.35)
	<i>cop_atl1</i>	1.300	(1.73)	0.249	(0.31)	0.092	(0.11)	0.451	(0.69)	1.823	(2.69)	1.631	(2.18)	0.337	(0.41)	1.549	(2.24)
	<i>gp_at</i>	1.165	(1.53)	0.191	(0.23)	0.216	(0.29)	0.390	(0.70)	1.785	(2.67)	1.623	(2.40)	0.621	(0.90)	1.594	(2.30)
	<i>opex_at</i>	0.399	(0.67)	0.239	(0.32)	0.405	(0.69)	0.066	(0.10)	0.268	(0.44)	0.108	(0.17)	-0.032	(0.05)	0.113	(0.18)
	<i>qmj_prof</i>	0.318	(0.44)	-0.861	(1.27)	-0.563	(0.89)	-0.645	(1.01)	0.998	(1.61)	0.892	(1.38)	-0.328	(0.52)	0.565	(0.81)
Seasonality	<i>pi_nix</i>	0.895	(1.44)	0.471	(0.72)	0.088	(0.14)	0.230	(0.36)	1.169	(1.80)	1.025	(1.71)	0.045	(0.07)	1.149	(1.79)
Size	<i>me</i>	2.856	(5.44)	2.521	(4.50)	3.021	(5.75)	2.405	(4.89)	2.391	(4.84)	2.463	(4.83)	2.440	(5.18)	2.318	(5.02)
Value	<i>at_me</i>	0.954	(1.49)	0.349	(0.48)	1.866	(2.24)	0.206	(0.28)	0.264	(0.50)	0.775	(1.25)	1.404	(1.71)	0.232	(0.40)
	<i>be_me</i>	1.157	(1.38)	0.862	(0.85)	2.304	(2.02)	0.523	(0.53)	0.336	(0.52)	0.784	(0.96)	1.679	(1.53)	0.542	(0.70)
	<i>div12m_me</i>	2.071	(2.65)	1.301	(1.61)	2.015	(2.35)	1.591	(1.71)	1.904	(2.39)	2.001	(2.42)	1.500	(1.74)	1.627	(2.02)
	<i>ebitda_mev</i>	1.332	(1.70)	0.225	(0.28)	1.087	(1.07)	0.139	(0.17)	1.371	(1.97)	1.621	(2.09)	1.320	(1.46)	1.282	(1.72)
	<i>eq_dur</i>	1.559	(1.88)	0.355	(0.43)	1.817	(2.13)	-0.012	(0.02)	1.176	(1.47)	1.160	(1.46)	1.285	(1.51)	0.940	(1.14)
	<i>eqnetis_at</i>	1.674	(1.68)	2.005	(1.82)	1.036	(0.99)	0.866	(0.83)	1.845	(1.74)	1.202	(1.08)	0.785	(0.78)	1.532	(1.35)
	<i>eqnpo_12m</i>	1.086	(1.58)	0.153	(0.23)	0.191	(0.29)	0.089	(0.15)	1.542	(2.44)	1.287	(2.01)	-0.153	(0.23)	1.082	(1.60)
	<i>netis_at</i>	0.778	(0.77)	0.499	(0.47)	0.718	(0.78)	0.547	(0.50)	1.081	(1.08)	0.642	(0.66)	0.053	(0.05)	1.001	(0.95)
	<i>ni_me</i>	1.679	(2.76)	1.043	(1.59)	1.738	(2.43)	1.075	(1.55)	1.694	(2.89)	1.825	(2.94)	1.515	(2.08)	1.406	(2.33)
	<i>ocf_me</i>	1.950	(3.04)	0.698	(1.09)	1.671	(2.43)	0.870	(1.35)	2.001	(3.01)	2.320	(3.31)	1.486	(1.81)	1.686	(2.50)
	<i>sale_me</i>	1.924	(3.10)	0.857	(1.40)	1.741	(2.90)	0.339	(0.61)	1.581	(2.63)	1.675	(2.80)	1.156	(1.86)	1.145	(1.88)

Table IA5: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Subsamples. Cont.

Panel C: Main Board Stocks

Cluster	Char	CAPM		IDX3-size-value		IDX3-size-prof		IDX4		FF3		Car4		Q4		FF5	
		α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$
Accruals	<i>oaccruals_at</i>	0.455	(1.10)	-0.032	(0.07)	0.274	(0.58)	0.020	(0.04)	0.441	(1.07)	0.301	(0.71)	0.253	(0.62)	0.166	(0.36)
Debt Issuance	<i>noa_at</i>	0.967	(2.21)	0.811	(1.81)	0.405	(0.97)	0.554	(1.18)	1.017	(2.25)	0.888	(1.98)	0.997	(2.33)	0.387	(0.86)
Investment	<i>mispricing_mgmt</i>	0.961	(2.69)	0.554	(1.34)	0.819	(2.06)	0.601	(1.43)	0.988	(2.74)	1.012	(2.78)	0.762	(1.96)	0.408	(1.06)
	<i>noa_gr1a</i>	1.504	(3.19)	1.348	(2.67)	1.122	(2.20)	1.307	(2.39)	1.517	(3.15)	1.354	(2.63)	1.492	(3.02)	0.755	(1.36)
Profit Growth	<i>dsale_dinv</i>	1.196	(2.23)	1.481	(2.67)	1.177	(2.09)	1.385	(2.46)	1.246	(2.30)	1.306	(2.45)	1.438	(2.71)	1.175	(2.05)
	<i>ocf_at_chg1</i>	0.621	(1.71)	0.502	(1.40)	0.463	(1.20)	0.386	(1.04)	0.511	(1.46)	0.305	(0.85)	0.517	(1.44)	0.419	(0.90)
Profitability	<i>ocf_at</i>	0.640	(1.24)	0.648	(1.25)	-0.205	(0.45)	0.328	(0.72)	1.032	(2.34)	0.815	(2.02)	0.766	(1.79)	0.039	(0.09)
Quality	<i>cop_at</i>	0.787	(1.41)	0.759	(1.32)	-0.170	(0.34)	0.214	(0.41)	1.115	(2.22)	0.907	(1.91)	0.829	(1.69)	0.218	(0.43)
	<i>cop_at1</i>	0.793	(1.36)	0.835	(1.41)	-0.070	(0.13)	0.384	(0.68)	1.134	(2.14)	0.969	(1.84)	0.906	(1.71)	0.306	(0.55)
	<i>gp_at</i>	1.210	(2.59)	1.457	(3.08)	0.239	(0.55)	0.895	(2.22)	1.551	(3.78)	1.325	(3.42)	1.449	(3.52)	0.580	(1.38)
	<i>opex_at</i>	0.374	(0.78)	0.253	(0.50)	-0.190	(0.41)	-0.103	(0.20)	0.444	(0.91)	0.218	(0.47)	0.224	(0.46)	-0.390	(0.84)
	<i>qmj_growth</i>	0.914	(2.44)	0.895	(2.17)	0.224	(0.66)	0.721	(1.69)	1.009	(2.85)	0.793	(2.26)	1.023	(2.81)	0.660	(1.63)
	<i>sale_bev</i>	0.850	(2.63)	0.987	(2.58)	0.167	(0.46)	0.584	(1.64)	1.141	(3.83)	1.156	(3.66)	0.974	(3.00)	0.489	(1.33)
Seasonality	<i>dbnetis_at</i>	0.384	(1.20)	0.097	(0.28)	0.275	(0.76)	0.229	(0.63)	0.455	(1.41)	0.423	(1.27)	0.377	(1.13)	0.124	(0.35)
	<i>pi_nix</i>	0.518	(1.35)	0.638	(1.70)	0.641	(1.77)	0.570	(1.53)	0.579	(1.52)	0.690	(1.79)	0.609	(1.64)	0.732	(2.10)
Size	<i>dolvol_126d</i>	0.898	(2.14)	0.580	(1.34)	1.188	(2.23)	0.600	(1.39)	0.561	(1.46)	0.491	(1.37)	0.434	(1.14)	0.672	(1.59)
	<i>me</i>	1.175	(3.24)	0.870	(2.43)	1.368	(3.99)	0.736	(2.60)	0.652	(2.51)	0.677	(2.50)	0.675	(2.37)	1.174	(3.36)
Value	<i>at_me</i>	0.605	(1.39)	-0.317	(0.79)	0.682	(1.51)	-0.251	(0.61)	0.435	(1.10)	0.645	(1.59)	0.481	(1.09)	0.826	(1.62)
	<i>be_me</i>	0.621	(1.38)	-0.051	(0.10)	1.396	(3.27)	-0.107	(0.33)	-0.170	(0.62)	-0.112	(0.37)	-0.320	(1.05)	0.604	(1.64)
	<i>ebitda_mev</i>	1.130	(2.45)	0.020	(0.05)	1.463	(3.13)	0.446	(1.10)	0.868	(1.97)	1.126	(2.50)	0.715	(1.62)	1.277	(2.63)
	<i>eq_dur</i>	1.108	(2.49)	0.273	(0.67)	1.501	(3.52)	0.460	(1.26)	0.620	(1.62)	0.715	(1.89)	0.586	(1.52)	1.091	(2.75)
	<i>eqnetis_at</i>	1.475	(2.65)	1.442	(2.44)	1.443	(2.37)	1.084	(1.89)	1.465	(2.65)	1.329	(2.43)	1.404	(2.50)	1.456	(2.53)
	<i>eqpo_me</i>	0.115	(0.10)	-2.568	(1.40)	1.403	(1.00)	-5.430	(1.00)	-1.216	(1.10)	-1.074	(0.78)	5.340	(2.71)	3.376	(2.82)
	<i>fcf_me</i>	1.519	(2.75)	1.191	(2.01)	0.867	(1.40)	1.112	(1.80)	1.613	(3.01)	1.382	(2.39)	1.480	(2.53)	0.656	(0.99)
	<i>ival_me</i>	0.369	(0.73)	-0.714	(1.38)	0.860	(1.67)	-0.271	(0.59)	0.027	(0.06)	0.345	(0.79)	-0.083	(0.19)	0.750	(1.47)
	<i>netis_at</i>	0.780	(1.42)	0.607	(1.10)	1.064	(1.69)	0.986	(1.69)	0.871	(1.59)	0.731	(1.29)	0.650	(1.19)	-0.006	(0.01)
	<i>ni_me</i>	0.889	(1.84)	0.364	(0.86)	0.607	(1.29)	0.131	(0.27)	0.868	(1.88)	0.715	(1.61)	0.669	(1.49)	0.542	(1.17)
	<i>ocf_me</i>	1.566	(3.10)	0.442	(0.89)	1.225	(2.38)	0.750	(1.44)	1.594	(3.15)	1.576	(3.14)	1.169	(2.32)	0.894	(1.98)
	<i>sale_me</i>	1.176	(2.36)	0.271	(0.64)	1.631	(3.23)	0.453	(1.19)	0.600	(1.53)	0.600	(1.44)	0.316	(0.81)	1.076	(2.42)

Table IA4: Alphas of Long-Short Portfolios relative to Different Asset Pricing Model – Subsamples. Cont.

Panel D: non-Main Board Stocks

Cluster	Char	CAPM		IDX3-size-value		IDX3-size-prof		IDX4		FF3		Car4		Q4		FF5	
		α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$	α_{LS}	$t(\alpha_{LS})$
Profit Growth	<i>dsale_dinv</i>	0.635	(0.84)	0.742	(0.86)	1.296	(2.02)	0.642	(1.04)	0.466	(0.62)	0.814	(1.05)	1.017	(1.45)	0.513	(0.71)
	<i>tax_gr1a</i>	1.184	(1.53)	1.293	(1.38)	1.173	(1.34)	1.286	(1.69)	1.149	(1.51)	1.664	(2.16)	1.194	(1.12)	1.394	(1.82)
Profitability	<i>ope_be</i>	0.635	(0.97)	-0.620	(0.95)	-0.004	(0.01)	-0.268	(0.45)	1.119	(1.88)	1.188	(1.93)	-0.104	(0.17)	0.766	(1.18)
	<i>ope_bell</i>	0.656	(1.01)	-0.437	(0.67)	0.040	(0.07)	-0.206	(0.35)	1.150	(1.97)	1.257	(2.08)	0.004	(0.01)	0.742	(1.18)
Size Value	<i>me</i>	2.856	(5.44)	2.521	(4.50)	3.021	(5.75)	2.405	(4.89)	2.391	(4.84)	2.463	(4.83)	2.440	(5.18)	2.318	(5.02)
	<i>at_me</i>	0.954	(1.49)	0.349	(0.48)	1.866	(2.24)	0.206	(0.28)	0.264	(0.50)	0.775	(1.25)	1.404	(1.71)	0.232	(0.40)
	<i>be_me</i>	1.157	(1.38)	0.862	(0.85)	2.304	(2.02)	0.523	(0.53)	0.336	(0.52)	0.784	(0.96)	1.679	(1.53)	0.542	(0.70)
	<i>div12m_me</i>	2.071	(2.65)	1.301	(1.61)	2.015	(2.35)	1.591	(1.71)	1.904	(2.39)	2.001	(2.42)	1.500	(1.74)	1.627	(2.02)
	<i>ebitda_mev</i>	1.332	(1.70)	0.225	(0.28)	1.087	(1.07)	0.139	(0.17)	1.371	(1.97)	1.621	(2.09)	1.320	(1.46)	1.282	(1.72)
	<i>eqnpo_12m</i>	1.086	(1.58)	0.153	(0.23)	0.191	(0.29)	0.089	(0.15)	1.542	(2.44)	1.287	(2.01)	-0.153	(0.23)	1.082	(1.60)
	<i>eqnpo_me</i>	2.494	(4.12)	1.370	(2.59)	1.806	(2.69)	0.401	(0.53)	2.424	(4.03)	2.560	(3.91)	1.952	(3.10)	2.292	(3.69)
	<i>fcf_me</i>	1.554	(3.00)	0.786	(1.59)	0.809	(1.36)	0.838	(1.46)	1.833	(3.61)	1.291	(2.38)	0.487	(0.85)	1.547	(2.65)
	<i>ival_me</i>	1.933	(2.09)	0.834	(0.83)	1.505	(1.60)	0.628	(0.63)	1.772	(1.75)	1.302	(1.24)	1.722	(1.74)	1.373	(1.35)
	<i>netis_at</i>	0.778	(0.77)	0.499	(0.47)	0.718	(0.78)	0.547	(0.50)	1.081	(1.08)	0.642	(0.66)	0.053	(0.05)	1.001	(0.95)
	<i>ni_me</i>	1.679	(2.76)	1.043	(1.59)	1.738	(2.43)	1.075	(1.55)	1.694	(2.89)	1.825	(2.94)	1.515	(2.08)	1.406	(2.33)
	<i>ocf_me</i>	1.950	(3.04)	0.698	(1.09)	1.671	(2.43)	0.870	(1.35)	2.001	(3.01)	2.320	(3.31)	1.486	(1.81)	1.686	(2.50)
	<i>sale_me</i>	1.924	(3.10)	0.857	(1.40)	1.741	(2.90)	0.339	(0.61)	1.581	(2.63)	1.675	(2.80)	1.156	(1.86)	1.145	(1.88)